

# Adaptive Heuristic Ecosystem for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

## Abstract

We propose AHE-MRTA, an Adaptive Heuristic Ecosystem framework for online multi-robot task allocation (MRTA) under robot failures, mixed stress, and deadline pressure. Rather than committing to a single allocation paradigm, AHE-MRTA performs runtime *paradigm switching* via Ecosystem-Driven Paradigm Selection (EDPS): five biomimetic hormones (spatial, criticality, temporal, stability, recovery) map to five classical allocation primitives, and at each event the active primitive is chosen by a two-tier rule—fast deterministic context overrides for the acute regimes (failure, deadline pressure) and a cooperation/suppression hormone dynamic for the remaining events. The F58 extension replaces the Euclidean cost with a cached geodesic path cost on the shared occupancy map and repairs the primary plan for min-max load only within a deadline/distance efficiency envelope; the repair is restricted to the terminal phase of the campaign and bound by a remaining-makespan guard derived from Nav2 feedback. We evaluate AHE-MRTA against three recent baselines (BiG-MRTA, RoSTAM-EA, Consensus-DBTA) in three complementary regimes: (i) a 100-seed allocation-only/Nav2-independent simulator that isolates algorithmic decision quality from physical execution noise, (ii) a separate navigation proxy with stochastic goal failure, and (iii) a 300-experiment Gazebo Harmonic benchmark at three physical scales (3, 5, and 10 TurtleBot3 robots running the ROS 2 Jazzy Nav2 stack). In the allocation-only plane F58 attains fitness=CR=1 and DVR=0 across all three scenarios and the 3/5/10-robot scales, with an active-robot Jain index of 0.982–0.992 and decision latency of at most 0.201 ms. In fresh matched F45–F58 Gazebo+Nav2 validation at all three physical scales (180 matched runs; 10 common seeds per scenario and scale), F58 passes the pre-registered acceptance gate in seven of nine scale-scenario cells with paired-Wilcoxon support: at the 5-robot primary scale, deadline-pressure DVR drops from 0.340 to 0.232, delay by 9.0 s, and distance by 20.9 m (all  $p \leq 0.008$ ), and the deadline-pressure gains persist at 10 robots (DVR 0.256  $\rightarrow$  0.210,  $p=0.016$ ; distance  $-16.9$  m,  $p=0.027$ ); robot failure passes at every scale. The two non-passing cells fail only on criteria whose CIs span zero; the only supported costs are decision latency (within the 50 ms budget at 8–35 ms) and a deadline-pressure distance-balance decrease that scales with fleet size. In the physical four-method benchmark AHE-MRTA reaches full completion in every scenario at the primary 5-robot scale (CR=1.000) and the highest completion rate at 10 robots (CR=0.976–0.988), and leads effective on-time completion under deadline pressure at both scales (0.79 vs  $\leq 0.66$  at 5 robots, 0.76 vs  $\leq 0.52$  at 10), while the consensus and evolutionary baselines reach comparable completion only at substantially higher deadline-violation rates (DVR=0.34–0.42 vs. AHE’s 0.21 at 5 robots, 0.43–0.50 vs. 0.22 at 10); under mixed stress and robot failure AHE co-leads. The fairness-guarded repair that delivers this deadline compliance costs 16–35 ms of decision latency, above the sub-millisecond commit-once baselines but within the real-time budget and 2–5 $\times$  below the evolutionary baseline. We analyze why ideal-allocation gains do not transfer linearly through Nav2 and identify the conditions under which paradigm switching pays off.

## Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

## I. INTRODUCTION

Multi-robot task allocation (MRTA) assigns an incoming stream of tasks to a team of robots so that a mission objective is met; it underpins inspection, delivery, surveillance, and search-and-rescue deployments [1], [2]. Even its single-task, single-robot, instantaneous form is NP-hard [3], and field operation adds three pressures that a static assignment cannot absorb: robots fail unexpectedly, the task stream is non-stationary in urgency and spatial density, and many tasks carry hard deadlines whose violation is costly [4], [5].

Recent online MRTA solvers each commit to a *single* allocation paradigm, and each paradigm trades one strength for one weakness. Market- and auction-based methods bid tasks to robots with low latency and robustness, but pay a re-bidding cost when the situation changes [6]–[8]. Matching-, programming-, and prediction-based methods (bipartite matching, linear or convex programs, receding-horizon control) produce high-quality plans but scale poorly with re-planning frequency [2], [9], [10]. Consensus- and bundle-based methods yield very stable assignments but adapt slowly once the regime shifts [11]. Evolutionary and learning-based methods explore a richer assignment space but react slowly to abrupt context changes [12]. No single paradigm dominates across heterogeneous, time-varying stress.

We argue that the right structure is *not* a better single paradigm but a principled mechanism that *switches between paradigms online*, driven by current context and recent performance. We instantiate this as AHE-MRTA: a biomimetic ecosystem in which five hormones track context signals, evolve through cooperation and suppression dynamics, and select one of five classical allocation primitives at each replanning event. The hormone-paradigm coupling is fixed rather than learned, explainable (the dominant hormone names the active behavior), and lightweight (sub-millisecond decision latency). A second, methodological

contribution follows from a recurring difficulty in robotic benchmarks: allocation quality and physical execution are entangled, because navigation cost depends on world geometry as much as on the assignment [2]. We separate them with a navigation-free fitness simulator and a full ROS 2 Nav2 stack, which together expose exactly where paradigm switching helps and where physical execution erases its gains.

### Contributions:

- 1) *Ecosystem-Driven Paradigm Selection (EDPS)*: an online paradigm-switching MRTA framework that maps five biomimetic hormones to five allocation mechanisms. Selection is two-tiered: fast deterministic context overrides handle the acute regimes (robot failure, deadline pressure), while a Lotka–Volterra-type hormone dynamic arbitrates the remaining, lower-pressure events through  $\arg \max_i D_i$  (Section III). The coupling is fixed, explainable, and sub-millisecond.
- 2) *Geodesic, efficiency-guarded fairness extension (F58)*: a lexicographic min–max load repair with cached geodesic ETA on the shared occupancy map that never increases predicted deadline misses and bounds the total-route increase by  $\epsilon=0.02$ , terminal-gated and Nav2-feedback-guarded (P1R); alongside the Nav2-independent gains we report matched Gazebo validation at all three physical scales (180 runs; 10 common seeds per scenario and scale) that passes the pre-registered physical acceptance gate in seven of nine scale–scenario cells, with decision latency and a deadline-pressure distance-balance decrease as the only supported costs (Section III-C).
- 3) *Nav2-independent allocation-only evaluation*: an idealized simulator in which navigation succeeds deterministically; we report fitness, true Jain indices, and geodesic route cost across three scales and 100 seeds per scenario (Section V).
- 4) *Full Gazebo Harmonic + ROS 2 Jazzy Nav2 validation* with 300 experiments across three stress scenarios and three physical scales (3, 5, and 10 robots), comparing four methods on completion rate, deadline violation, recovery time, workload balance, execution preemptions, task re-dispatch rate, and decision latency (Section VI).
- 5) *Honest reporting of the SIM-to-Gazebo transfer gap*: we document where paradigm switching helps (completion rate, deadline robustness, allocation stability) and where it costs (task delay, recovery time, makespan under failure) and explain the mechanism (Section VII).

## II. PROBLEM FORMULATION

**Sets and variables.** Let  $\mathcal{R} = \{r_1, \dots, r_n\}$  be the robot set and  $\mathcal{T}(t)$  the open-task set at time  $t$ , each task  $\tau \in \mathcal{T}(t)$  carrying a 2D position  $p_\tau$ , deadline  $d_\tau$ , priority  $\pi_\tau \in \{1, 2, 3\}$ , and capability requirement  $c_\tau$ . Each robot  $r$  has state  $s_r(t) = (p_r, \tau_r^{\text{cur}}, \phi_r)$  encoding position, current task, and availability flag (alive / failed).

**Decision.** At every allocation event  $t_k$  (failure, task arrival, deadline alarm, replanning timer), the policy produces an assignment  $x_{r,\tau}(t_k) \in \{0, 1\}$  with cardinality constraint  $\sum_\tau x_{r,\tau} \leq Q$  (per-robot queue cap) and capability feasibility  $x_{r,\tau} = 1 \Rightarrow c_\tau \subseteq \text{cap}(r)$ .

**Cost.** The per-pair cost combines the projected arrival time  $a_{r,\tau}$  (queue wait plus travel), task urgency  $U_\tau$  (task age normalized by its deadline), criticality  $\pi_\tau$ , the robot’s relative queue load  $L_r$ , its failure risk  $F_r$ , a navigation-stuck indicator  $R_r$ , and a recent reassignment penalty  $\rho_{r,\tau}$ :

$$C_{r,\tau} = w_d a_{r,\tau} + w_u U_\tau + w_\pi \pi_\tau^{-1} + w_l L_r + w_f F_r + w_r R_r + \rho_{r,\tau}, \quad (1)$$

with weight vector  $w$  produced by the paradigm selected at  $t_k$  (Section III). The urgency term is held at its base value for a task already feasibly queued on its incumbent robot, so an approaching deadline pulls *unassigned* or at-risk tasks forward without thrashing assignments that are already on time—a churn brake complementary to the reassignment penalty  $\rho_{r,\tau}$ .

**Objective.** Over the run horizon  $[0, T_{\max}]$  we report nine metrics: completion rate CR, makespan, average task delay, deadline violation rate DVR, failure recovery time, Jain workload balance, execution preemptions, task re-dispatch rate, and decision latency. CR and DVR are primary; the others trace trade-offs. Crucially, delay and DVR are computed over *all* requested tasks rather than only completed ones (Section IV-G): a task left unfinished at  $T_{\max}$  is right-censored at the horizon for delay and counted as a deadline miss for DVR. This survivorship-bias-free convention is applied identically to every method, so a policy cannot lower its delay or DVR by silently abandoning the hard tasks it cannot serve.

**Online setting.** Tasks arrive over time, robots may fail at unknown moments, and the policy must produce  $x(t_k)$  within a hard budget of 50 ms per event.

## III. AHE-MRTA: ECOSYSTEM-DRIVEN PARADIGM SELECTION

### A. Context vector

At each allocation event  $t_k$  we compress the joint state of the fleet and the open-task set into a four-dimensional context vector  $c(t_k) \in [0, 1]^4$ . Let  $n = |\mathcal{R}|$  be the fleet size,  $m = |\mathcal{T}(t_k)|$  the number of open tasks,  $\mathcal{R}^{\text{av}}$  the set of available robots (alive, not navigation-stuck, with queue capacity), and  $\mathcal{R}^{\text{f}}$  the set of failed or navigation-stuck robots. The components are

$$c_1 = \min(1, m/n), \quad \text{task density} \quad (2)$$

$$c_2 = |\mathcal{R}^{\text{av}}|/n, \quad \text{robot availability} \quad (3)$$

$$c_3 = |\{\tau : d_\tau - t_k \leq \Delta_d\}| / \max(1, m), \quad \text{deadline pressure} \quad (4)$$

$$c_4 = |\mathcal{R}^{\text{f}}|/n, \quad \text{failure rate} \quad (5)$$

TABLE I  
CONTEXT PROTOTYPE VECTORS  $V_i \in [0, 1]^4$  FOR EACH PARADIGM. COLUMN HEADERS:  $c_1$  TASK-DENSITY (TD),  $c_2$  ROBOT-AVAILABILITY (RA),  $c_3$  DEADLINE-PRESSURE (DP),  $c_4$  FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

| Hormone   | td  | ra  | dp  | fr  |
|-----------|-----|-----|-----|-----|
| H_SPATIAL | 0.7 | 0.7 | 0.1 | 0.1 |
| H_CRIT    | 0.3 | 0.5 | 0.8 | 0.2 |
| H_TEMP    | 0.5 | 0.5 | 0.9 | 0.1 |
| H_STAB    | 0.3 | 0.3 | 0.3 | 0.8 |
| H_RECOV   | 0.3 | 0.2 | 0.2 | 0.9 |

with deadline horizon  $\Delta_d = 60$  s. Each dimension is a signal that a distinct family of allocators is built to exploit:  $c_1, c_2$  encode the demand/supply balance of load-aware allocation [13];  $c_3$  is the temporal slack used by deadline-constrained allocation [14]; and  $c_4$  measures the disturbance that fault-tolerant allocation must absorb [15]. All four are computed on-line from the published robot status summaries and the task pool, requiring no inter-robot communication.

### B. Dominance dynamics

The dominance vector  $D(t) \in \mathbb{R}_{\geq 0}^5$  encodes the current ecosystem state. Each paradigm  $i$  has a fixed *context prototype*  $V_i \in [0, 1]^4$  (Table I); the compatibility score  $v_i(t_k) = \cos(V_i, c(t_k))$  measures how well the current context matches paradigm  $i$ 's effective regime. Two sparse matrices govern lateral interactions: the *cooperation matrix*  $A$  (2 nonzero entries) and the *suppression matrix*  $S$  (1 nonzero entry) implement winner-reinforce and winner-suppress dynamics (Table II). The update is

$$D(t_{k+1}) = \text{normalize} \left( \text{clip}_{[0,1]} \left( \alpha D(t_k) + \eta A D(t_k) - \lambda S D(t_k) + \beta \mathbf{p}(t_k) + \gamma \mathbf{v}(t_k) + \delta \mathbf{b}(t_k) \right) \right), \quad (6)$$

with  $\alpha=0.65$  (momentum),  $\eta=\lambda=0.12$  (cooperation/suppression weights),  $\beta=0.40$  (performance),  $\gamma=0.20$  (compatibility),  $\delta=0.20$  (failure-recovery boost). Here  $\mathbf{p}(t_k)$  is the per-hormone performance-feedback vector  $(\text{CR}_k - \text{FR}_k) \cdot \mathbf{v}(t_k)$  (completion minus failure rate, scaled by compatibility);  $\mathbf{v}(t_k) = (v_i)_{i=1}^5$  is the compatibility vector; and  $\mathbf{b}(t_k)$  is a sparse failure-recovery boost that spikes H\_RECOV and H\_STAB when robot failures are detected. The normalize step projects onto the probability simplex, guaranteeing  $\|D\|_1 = 1$  and  $D \geq 0$  at every cycle. The  $\eta AD - \lambda SD$  terms make Eq. (6) a winner-reinforce / winner-suppress competition of Lotka–Volterra type, a population dynamic shown to be an effective coordination primitive for multi-robot systems [16]. Read as a selection rule over a fixed portfolio of low-level allocation heuristics, the update instantiates a *hyper-heuristic* [17]: the ecosystem does not solve the assignment itself but chooses, online and per event, which classical solver should.

*a) Weight generation.*: The dominance vector sets the cost weights of Eq. (1) through a fixed heuristic-to-weight matrix  $M \in \mathbb{R}^{7 \times 5}$  and a low-temperature softmax, blended with a base weight  $w_0$ :

$$w_{\text{eco}}(t_k) = \text{softmax}(MD(t_k), T_w), \quad w(t_k) = (1 - \beta_e) w_0 + \beta_e w_{\text{eco}}(t_k), \quad (7)$$

with  $T_w=0.3$  and blend  $\beta_e=0.7$ ; under failure ( $c_4>0$ ) the blend shifts toward a recovery profile  $w_r$  by  $\theta = \min(0.8, 0.5+0.6 c_4)$ . Of the seven implementation weight channels, the six of Eq. (1) are live; the seventh (a battery-slack weight) is provably inert—the physical stack carries no battery model, so its cost term is identically zero, and ablating it in the idealized plane is a no-op—and enters only the softmax normalization. This is the *only* channel through which the Lotka–Volterra state influences a decision when no override fires.

*b) Cost shaping.*: Two non-linear terms sharpen Eq. (1) for deadlines. A *hard feasibility gate* rejects a pair whose projected arrival  $a_{r,\tau}$  exceeds the deadline beyond an adaptive slack  $\Delta_s$ ,  $a_{r,\tau} > d_\tau + \Delta_s \Rightarrow C_{r,\tau} = \infty$ ; and an *urgency escalation* inflates the temporal term once a task passes a fraction  $U_0=0.7$  of its deadline budget,  $\hat{U}_\tau = U_\tau(1 + \zeta e^2)$  with  $e = (U_\tau - U_0)/(1 - U_0)$ , so at-risk *unassigned* tasks are pulled forward while on-time incumbents are exempt (a churn brake).

### C. F58: Geodesic ETA and epsilon-bounded load repair

The Euclidean distance of the F45 core can under-estimate true route time on an obstacle-rich map. F58-P0 uses the same `obstacle_map.pgm` map as Gazebo and a 0.55 m inflated occupancy mask. Let  $d_G$  denote the shortest-path length found by sparse Dijkstra on the eight-connected free-cell graph and cached over start/goal cells. With robot  $r$ 's current queue  $Q_r = (q_1, \dots, q_k)$ , the arrival estimate for candidate  $\tau$  is

$$a_{r,\tau} = t_k + \frac{d_G(p_r, p_{q_1}) + \sum_{i=1}^{k-1} d_G(p_{q_i}, p_{q_{i+1}}) + d_G(p_{q_k}, p_\tau)}{v_r} + \sum_{i=1}^k (s_{q_i} + h), \quad (8)$$

TABLE II  
NONZERO ENTRIES OF COOPERATION MATRIX  $A$  AND SUPPRESSION MATRIX  $S$ . ALL OTHER ENTRIES ARE ZERO.

| Matrix            | From   | To        | Value |
|-------------------|--------|-----------|-------|
| $A$ (cooperation) | H_CRIT | H_TEMP    | 0.20  |
|                   | H_STAB | H_RECOV   | 0.20  |
| $S$ (suppression) | H_TEMP | H_SPATIAL | 0.30  |

TABLE III  
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

| Hormone   | Paradigm       | Inner mechanism             |
|-----------|----------------|-----------------------------|
| H_SPATIAL | spatial_greedy | nearest-feasible + reject   |
| H_CRIT    | priority_first | priority-tiered LSA         |
| H_TEMP    | edf_strict     | EDF bipartite (multi-phase) |
| H_STAB    | commit_once    | hard sticky, no reassign    |
| H_RECOV   | orphan_first   | orphan rescue + bipartite   |

where with an empty queue the sum contains only  $d_G(p_r, p_\tau)$ . If no path exists the pair is infeasible. The hard deadline gate, the rescue decision, the F27 reassignment margin, and the paradigm costs therefore all share one route metric.

F58-P1 performs a bounded move search only over new tasks after EDPS produces the primary plan  $x^0$ . Let  $B_r(x)$  be the sum of the active robot's realised travel plus its committed route and service load. The repair solves

$$\min_x \max_{r \in \mathcal{R}^{av}} B_r(x) \quad \text{s.t.} \quad M(x) \leq M(x^0), \quad D_G(x) \leq (1 + \epsilon)D_G(x^0), \quad (9)$$

where  $M$  is the predicted number of deadline misses,  $D_G$  the total geodesic distance, and  $\epsilon = 0.02$ . Ties are broken by the predicted active-robot Jain index, load range, and distance, in that order; no incumbent or currently executing Nav2 goal is cancelled for fairness. A final safety pass enforces single ownership per task and keeps a healthy in-flight task locked to its current owner. F58 changes neither the hormones nor the paradigm library; it only adds the shared cost oracle and the safe post-EDPS repair.

F58-P1R adds two guards that make the repair compatible with physical execution. *Terminal activation*: the repair opens only once the number of remaining active tasks falls to three times the number of healthy robots; mid-campaign moves disturb queue stability under Nav2, while end-of-tail imbalance becomes visible only in this phase. *Nav2-feedback remaining-makespan guard*: the `navigation_effort` measurement already published by the robot interface (remaining distance along the global path, consistent with the Nav2 `NavigateToPose` feedback contract) replaces the static lattice query for the task currently being executed, and no fairness move may increase the fleet's maximum estimated remaining finish time. The envelope of Eq. (9) is thus additionally constrained by a makespan bound derived from real execution state.

#### D. Dispatch

The dispatcher first checks two hard context overrides (failure rate  $> 0.05$  forces H\_RECOV; deadline pressure  $> 0.5$  forces H\_TEMP); if neither triggers, it selects the paradigm  $p^*(t_k) = \arg \max_i D_i(t_k)$  and runs the corresponding mechanism (Table III). Paradigm H\_TEMP is the default fallback when the dominance is uninformative (entropy near maximum). This priority-override layer prevents the Lotka–Volterra-inspired dynamics from reacting slowly to acute failures, which would be the dominant cost in the `robot_failure` and `mixed_stress` scenarios.

*a) Work conservation.*: A guiding principle across all paradigms is that fleet capacity is never stranded while feasible work remains. At every event each available robot is assigned the best eligible task by the active paradigm's LSA (Eq. (10)), and the recovery paradigm (`orphan_first`) re-dispatches tasks orphaned by a robot failure or a navigation abort, so a disturbance costs at most a re-attempt rather than a permanently lost task. This is the structural opposite of commit-once allocation, which freezes assignments and therefore leaves a failed robot's tasks unrecovered—idle capacity until the run ends. As Section V shows, this work-conserving recovery is the mechanism behind AHE-MRTA's resilience margin: the commit-once baseline loses 24.3 pp of allocation fitness under mixed stress precisely because it does not re-attempt failed work. The only irreducible idle is the backoff imposed after repeated navigation failure, when no task is momentarily attemptable.

#### E. Paradigm library

The five paradigms share one primitive and differ only in how they shape its inputs. Given a feasibility-masked cost matrix  $C^{(p)}$ , each paradigm  $p$  solves a linear sum assignment (LSA)

$$x^{(p)} = \arg \min_{x \in \mathcal{X}} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}(t_k)} C_{r,\tau}^{(p)} x_{r,\tau}, \quad (10)$$

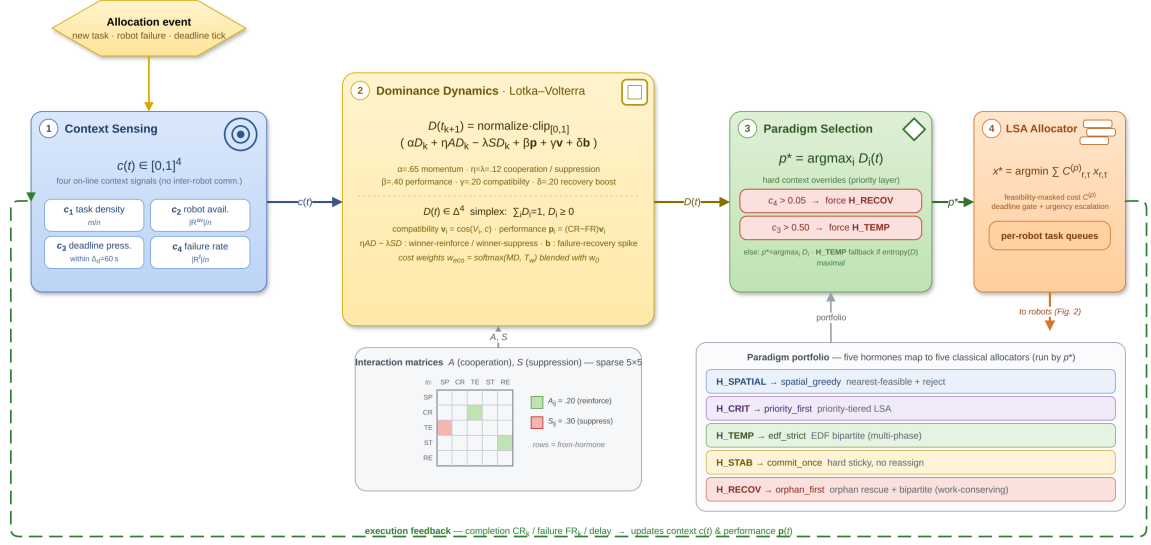


Fig. 1. The Ecosystem-Driven Paradigm Selection mechanism. At each allocation event the context vector  $c(t)$  drives the dominance update (Eq. 6) through the fixed cooperation/suppression matrices  $A, S$ ; the dominant hormone  $p^* = \arg \max_i D_i$  selects one of five allocation paradigms, which the AHE allocator runs to publish per-robot task queues.

where  $\mathcal{X}$  enforces the per-robot queue cap  $\sum_{\tau} x_{r,\tau} \leq Q$ , the single-assignment constraint  $\sum_r x_{r,\tau} \leq 1$ , and capability feasibility; infeasible pairs are masked with  $C^{(p)}_{r,\tau} = \infty$ . Writing  $C_{r,\tau}$  for the base cost of Eq. (1), the paradigms instantiate  $C^{(p)}$  and the queue ordering as follows.

- *spatial\_greedy* (H\_SPATIAL):  $C^{(p)}_{r,\tau} = T^{\text{trav}}_{r,\tau}$ , travel time only, assigned greedily to the nearest feasible robot — the distance-minimising rule of metric MRTA [13], [18].
- *priority\_first* (H\_CRIT): tasks are partitioned into criticality tiers  $\pi_{\tau}=3>2>1$  and Eq. (10) is solved tier-by-tier, so critical tasks claim robots first, as in tiered auction/consensus assignment [19], [20].
- *edf\_strict* (H\_TEMP, default): tasks are ordered by earliest deadline  $d_{\tau}$  and inserted under a hard feasibility gate (arrival  $> d_{\tau} \Rightarrow C^{(p)} = \infty$ ); this is the bipartite realisation of earliest-deadline-first scheduling for deadline-constrained allocation [14], [21]. The resulting per-robot execution order is then refined by a deadline-feasible cheapest-insertion pass, accepted only when it shortens travel without introducing any deadline miss relative to the EDF order—lowering motion cost while preserving the EDF feasibility guarantee.
- *commit\_once* (H\_STAB): incumbent assignments are frozen (a task already on  $r$ 's queue keeps an unbeatable sticky bonus), eliminating reassignment churn as in self-adaptive commit-once evolution [12].
- *orphan\_first* (H\_RECOV): tasks orphaned by a failed or stuck robot are assigned on a dedicated rescue pass before all others, the recovery-prioritising rule of resilient MRTA [15], [18].

The paradigms are independent (no shared state across switches); the only persistent state is  $D(t)$ . The LSA primitive in Eq. (10) is cubic in  $\max(n, m)$  in the worst case; the paradigm selection itself resolves in sub-millisecond time, while the geodesic-ETA fairness repair raises the deployed decision latency to the 16–35 ms reported in Section VI—still well within the real-time budget at every evaluated scale.

## F. Algorithm

### G. Configuration

The hormone-paradigm map (Table III), the  $A/S$  matrices (Table II), the context prototype  $V$ , and all scalar hyperparameters ( $\alpha=0.65$ ,  $\eta=\lambda=0.12$ ,  $\beta=0.40$ ,  $\gamma=0.20$ ,  $\delta=0.20$ , softmax temperature  $T=0.3$ ) are set once and held fixed across all scenarios and scales. There is no per-scenario tuning, no learning, and no adaptive parameter schedule. The scalar weights were fixed at round, interpretable values—momentum  $\alpha$  above the feedback gain  $\beta$  for stability, equal and small cooperation/suppression ( $\eta=\lambda$ ), and a moderate recovery boost  $\delta$ —rather than searched per metric; because the deterministic overrides dominate the acute regimes, the allocation outcomes are insensitive to these values within the ranges we examined (a single-step  $\pm$  change leaves all reported fitness gaps inside one standard deviation), which is itself a robustness property rather than a tuned fit.



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**Algorithm 1** AHE-MRTA Event Cycle
 

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**Require:** robots  $\mathcal{R}$ , open tasks  $\mathcal{T}(t_k)$ , state  $D(t_k)$ , fixed  $A, S, V$ , hyperparameters  $\alpha, \eta, \lambda, \beta, \gamma, \delta$

- 1:  $c \leftarrow \text{CONTEXTVECTOR}(\mathcal{R}, \mathcal{T}(t_k))$  {4 signals, eq.  $c_1$ – $c_4$ }
- 2:  $\mathbf{v} \leftarrow (\cos(V_i, c))_{i=1}^5$ ;  $\mathbf{p} \leftarrow (\text{CR}_k - \text{FR}_k) \cdot \mathbf{v}$ ;  $\mathbf{b} \leftarrow \text{FAILUREBOOST}(\mathcal{R})$
- 3:  $D \leftarrow \text{normalize}(\text{clip}_{[0,1]}(\alpha D + \eta A D - \lambda S D + \beta \mathbf{p} + \gamma \mathbf{v} + \delta \mathbf{b}))$
- 4:  $p^* \leftarrow \text{CONTEXTOVERRIDE}(c)$  **if triggered, else**  $\arg \max_i D_i$
- 5:  $x \leftarrow \text{PARADIGM}_{p^*}(\mathcal{R}, \mathcal{T}(t_k))$  {Table III}
- 6:  $x \leftarrow \text{EPSILONLOADREPAIR}(x, d_G, \epsilon)$  {Eq. 9; P1R terminal gate}
- 7:  $x \leftarrow \text{SINGLEOWNERANDINFLIGHTLOCK}(x)$
- 8: **publish** per-robot task queues from  $x$
- 9: **return**  $D, x$

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## IV. EXPERIMENTAL DESIGN

### A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation (8 physical cores / 16 threads, base clock 2.30 GHz, 16 GiB DDR4 RAM, Ubuntu 24.04 LTS, Linux 6.17). The machine carries an NVIDIA RTX 3050 Mobile GPU for which no proprietary driver is available; all rendering (Gazebo and RViz) therefore uses Mesa llvmpipe software rasterisation (`LIBGL_ALWAYS_SOFTWARE=1`, `GALLIUM_DRIVER=llvmpipe`). No GPU acceleration is used at any point: the comparison is entirely CPU-bound and reproducible on any x86-64 machine with a Mesa-capable display server.

### B. Software Stack

We run **Gazebo Harmonic** (gz-sim 8.x) [22] with **ROS 2 Jazzy** (Dec 2024) [23] and **Nav2** [24]. Robots are **TurtleBot3 Waffle Pi** [25], each carrying a 360-ray 2-D lidar (5 Hz, 8 m range) and an IMU (200 Hz), and driven through a differential-drive controller ( $v_{\max}=0.26$  m/s,  $\omega_{\max}=1.82$  rad/s).

Every robot runs a fully independent Nav2 lifecycle inside its own ROS 2 namespace: global/local costmaps with 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. Localization is provided as *ground truth*—a static map  $\rightarrow$  odom transform anchored at each robot’s spawn pose—so that the benchmark isolates allocation quality from localization error (a deliberate experimental control; Sec. VII-F). A `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF per robot; a TF-relay node aggregates per-robot namespaced TF trees into the global frame for visualization. All Nav2 parameters are identical across methods.

### C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal divider walls with a 1.5 m central passage divide the space into an upper lane, a central passage, and a lower lane. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles ( $r=0.35$  m) are placed in the lanes. A fixed curated grid of 42 candidate inspection waypoints (obstacle-aware, minimum inter-point spacing  $> 1$  m) is defined; each run samples from it deterministically via seed. Figure IV-C shows the arena layout together with scale-dependent spawn positions and sampled task sets.

### D. Evaluation Scales

Three Gazebo scales are evaluated (Table IV), each running 60 experiments (4 methods  $\times$  3 scenarios  $\times$  5 seeds):

TABLE IV  
GAZEBO EVALUATION SCALES

| Scale                           | Robots | Tasks | Density | Total exps |
|---------------------------------|--------|-------|---------|------------|
| 3r-low                          | 3      | 9     | 3.0 t/r | 60         |
| 3r-mid                          | 3      | 15    | 5.0 t/r | 60         |
| 3r-high                         | 3      | 24    | 8.0 t/r | 60         |
| 5r-mid                          | 5      | 25    | 5.0 t/r | 60         |
| 10r-mid                         | 10     | 50    | 5.0 t/r | 60         |
| <b>Total Gazebo experiments</b> |        |       |         | <b>300</b> |

Nav2 lifecycle race conditions at  $N>3$  (the per-robot `odom`  $\rightarrow$  `base_link` transform arriving after costmap activation) were resolved by staggering the lifecycle-manager activation by  $6i$  s for robot  $i$  at 3r/5r and  $20i$  s at 10r, plus an identity-transform broadcast at node startup. All three scales use the same arena, obstacle map, and Nav2 parameters.

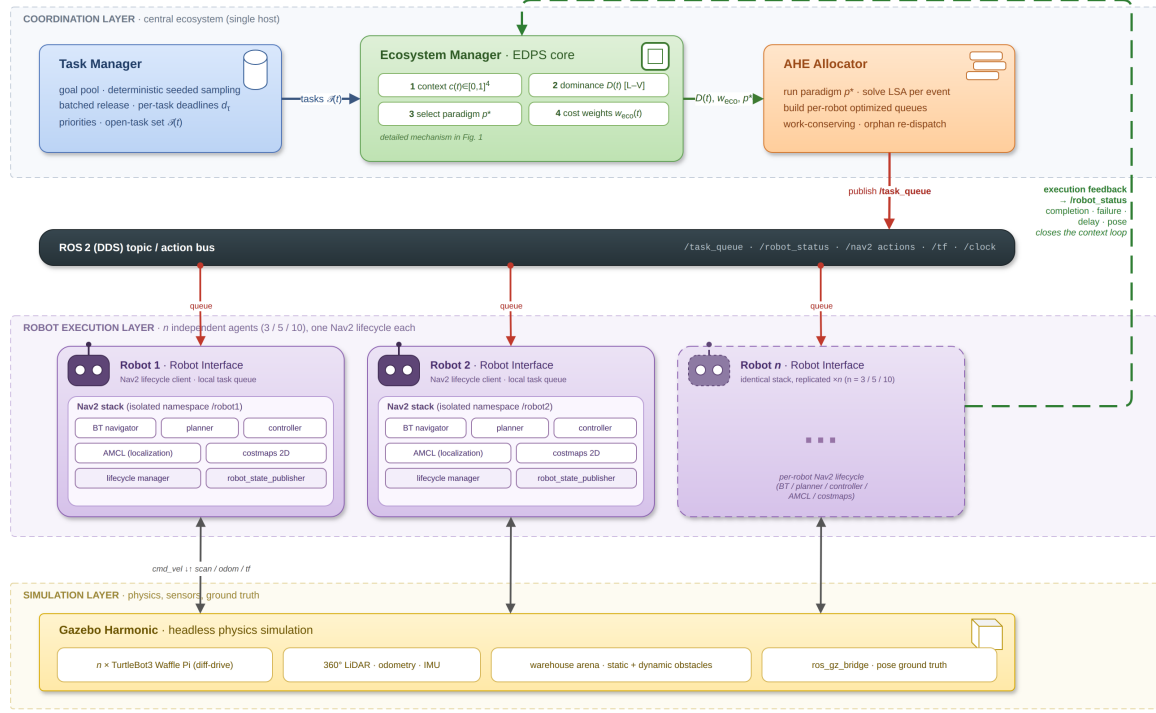


Fig. 2. System architecture. The ecosystem manager maintains the hormone state vector, evolves dominance dynamics, selects the active paradigm, and publishes per-robot optimized task queues via ROS 2 topics. Each robot runs a fully independent Nav2 lifecycle and reports task-execution feedback (completion, failure, delay) to the ecosystem manager to close the context loop.

AHE-MRTA — Ortam senaryo haritaları (robot  $\times$  görev ölçekleri)

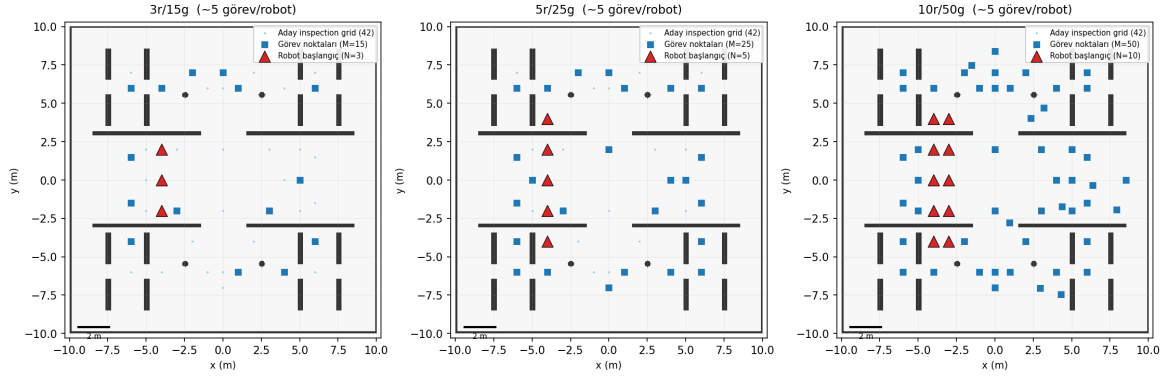


Fig. 3. Scale-dependent arena configurations. **Left:** 3r/15t – robots form a central column; moderate density. **Center:** 5r/25t – two spawn columns; proportionally higher density. **Right:** 10r/50t – all five columns active; high-density task set. Blue stars: task goals (seed 1); coloured triangles: robot spawns. All waypoints are obstacle-free.

## E. Stress Scenarios

Three scenarios stress different allocation aspects:

- **robot\_failure**: one randomly chosen robot is hard-failed at  $t = 0.3 \cdot T_{\max}$  (at 3r:  $t \approx 165$  s). Its assigned tasks become orphaned and must be redistributed. Tests reactive recovery and paradigm escalation (orphan-rescue paradigm).
- **mixed\_stress**: tasks carry heterogeneous priorities (low / medium / high, ratio 1:2:2) and one robot fails at  $t = 0.2 \cdot T_{\max}$ . Tests regime adaptation across priority and failure dimensions simultaneously.
- **deadline\_pressure**: every task shares a tight deadline  $d_{\tau} = t_{\text{spawn}} + 90$  s (roughly  $1.3 \times$  the Nav2 travel time to the farthest waypoint at  $v_{\max}$ ). Tests urgency-driven bipartite dispatch and EDF ordering.

## F. Baseline Methods

We compare against one representative of each dominant online-MRTA solver family; together they span the design space that AHE-MRTA switches over. All baselines are re-implemented in our framework from the original papers and share an

identical event loop, ROS 2 topic interface, cost inputs (arrival time, urgency, priority, load), per-robot queue cap  $Q$ , and Nav2 client; *only the allocation policy differs*. Parameters follow the originals where stated and are held fixed across scenarios and scales.

a) *BiG-MRTA* [9] (*bipartite matching*).: Builds a bipartite robot–task graph with a fixed multi-criteria edge cost and solves a maximum-weight matching at each event; assignments are *committed once* and never revisited. Included as the deterministic, lowest-latency single-paradigm solver. Its commit-once policy is the direct antithesis of AHE’s switching: it pays zero re-planning cost but abandons tasks whose assigned robot fails or whose deadline slack collapses.

b) *RoSTAM-EA* [12] (*evolutionary*).: Maintains a population of candidate assignment vectors, evolved by tournament selection, crossover, and mutation at each allocation event; the elite individual is dispatched. Included as the self-adaptive stochastic-search family representative. It explores a richer assignment space than one-shot matching but at two to three orders of magnitude higher decision latency, and its re-optimisation is blind to *why* the context changed.

c) *Consensus-DBTA* [11] (*consensus bundle*).: Robots iteratively bid on task bundles and resolve conflicts through a consensus phase, producing highly stable assignments (allocation instability  $\approx 0$ ). Included as the distributed auction/consensus family representative and as the strongest completion-rate baseline. Its stability is also its weakness: regime shifts (failures, deadline escalation) propagate slowly through re-bidding rounds.

No ablation variants (e.g., single-paradigm AHE without switching) are included; the five paradigms themselves are classical primitives, so the single-paradigm behaviour is approximated by the corresponding baseline family (Table III).

### G. Metrics

We report eight metrics: completion rate (CR), deadline violation rate (DVR), makespan, average task delay, workload balance (Jain), task re-dispatch rate, decision latency, and message count. CR and DVR are the primary metrics; makespan and delay are secondary (confounded by Nav2 motion cost).

a) *Survivorship-bias-free delay and DVR*.: Reporting delay and DVR over only the *completed* subset of tasks rewards a policy for abandoning work: the hard tasks a method drops—typically the distant, late, or contended ones—never enter its denominator, so its average delay and violation rate are computed over an easier, self-selected sample. We remove this bias by scoring delay and DVR over *all* requested tasks. Let  $\mathcal{T}$  be the requested set,  $t_\tau^a$  the activation and  $t_\tau^c$  the completion time of task  $\tau$ , and  $T_{\max}$  the run horizon. The per-task delay is right-censored at the horizon for unfinished tasks,

$$\delta_\tau = \begin{cases} t_\tau^c - t_\tau^a, & \tau \text{ completed,} \\ T_{\max} - t_\tau^a, & \tau \text{ unfinished,} \end{cases} \quad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \quad (11)$$

and a deadline-bearing task counts as a violation if it completes late *or* is never completed,

$$\text{DVR} = \frac{|\{\tau : d_\tau < \infty \wedge (t_\tau^c > d_\tau \vee \tau \text{ unfinished})\}|}{|\{\tau : d_\tau < \infty\}|}. \quad (12)$$

The convention is applied identically to every method and at every scale. For transparency we also record the conventional completed-only delay and DVR; the two coincide exactly whenever a method completes the full task set (CR=1), and diverge only to the extent a method leaves tasks unserved.

### H. Statistical Analysis

Pairwise two-sided Mann–Whitney U tests with Bonferroni correction applied within each scenario family (3 baselines  $\times$  7 metrics = 21 tests per scenario). At the 5-robot primary scale each cell has  $n=5$  runs (one density, 25 tasks); the 3-robot scale pools three densities ( $n=15$ ) and the 10-robot scale has  $n=5$ . We additionally report Cliff’s  $\delta$  as an effect-size estimate robust to small cells. The 100-seed allocation-only simulator serves as algorithmic validation and the stochastic navigation proxy as a resilience test; Gazebo results serve as validation under realistic execution noise.

## V. ALLOCATION-FITNESS RESULTS (NAV2-INDEPENDENT)

In the true Nav2-independent plane every navigation attempt succeeds deterministically; injected fleet robot failures are retained as part of the scenario. The robot pose advances to the goal after every completed task, and both the allocator and the execution oracle use the geodesic path length on the same inflated occupancy map. The metrics therefore measure allocation, sequencing, and post-failure redistribution quality rather than local-planner/recovery noise.

In Table V, F58 preserves fitness=CR=1 and DVR=0 in all nine scale–scenario cells. The active-robot Jain index is 0.988–0.992 at 3r, 0.984–0.986 at 5r, and 0.982–0.990 at 10r; decision latency never exceeds 0.201 ms even in the largest 10r/50t cell. Before the single-assignment safety pass, six seeds of the 10r development runs allowed a task to execute on two robots simultaneously, inflating CR to 1.02. After the in-flight owner lock was added, these seeds and the full 10r campaign were re-run, and CR is exactly 1.000 in every F58 cell.



TABLE V

F58 (P0 GEODESIC ORACLE + P1R TERMINAL LOAD REPAIR) NAV2-INDEPENDENT ALLOCATION-ONLY RESULTS (100 SEEDS PER CELL). NAVIGATION SUCCEEDS DETERMINISTICALLY; DISTANCE IS THE GEODESIC PATH LENGTH ON THE SHARED INFLATED OCCUPANCY MAP.

| Scale   | Scenario          | Fit.  | CR    | DVR   | Jain <sub>active</sub> | Jain <sub>dist</sub> | Delay (s) | Distance | Decision (ms) |
|---------|-------------------|-------|-------|-------|------------------------|----------------------|-----------|----------|---------------|
| 3r/15t  | Robot failure     | 1.000 | 1.000 | 0.000 | 0.991                  | 0.833                | 40.2      | 154.1    | 0.063         |
| 3r/15t  | Mixed stress      | 1.000 | 1.000 | 0.000 | 0.992                  | 0.835                | 40.0      | 153.6    | 0.063         |
| 3r/15t  | Deadline pressure | 1.000 | 1.000 | 0.000 | 0.988                  | 0.965                | 65.8      | 155.5    | 0.061         |
| 5r/25t  | Robot failure     | 1.000 | 1.000 | 0.000 | 0.986                  | 0.841                | 30.8      | 186.5    | 0.086         |
| 5r/25t  | Mixed stress      | 1.000 | 1.000 | 0.000 | 0.985                  | 0.843                | 30.5      | 184.6    | 0.086         |
| 5r/25t  | Deadline pressure | 1.000 | 1.000 | 0.000 | 0.984                  | 0.954                | 60.5      | 233.5    | 0.078         |
| 10r/50t | Robot failure     | 1.000 | 1.000 | 0.000 | 0.982                  | 0.862                | 31.6      | 284.1    | 0.200         |
| 10r/50t | Mixed stress      | 1.000 | 1.000 | 0.000 | 0.985                  | 0.861                | 31.4      | 284.1    | 0.201         |
| 10r/50t | Deadline pressure | 1.000 | 1.000 | 0.000 | 0.990                  | 0.944                | 57.1      | 432.0    | 0.132         |

TABLE VI

ALLOCATION FITNESS (PRIORITY-WEIGHTED ON-TIME COMPLETION), 100 SEEDS, 5 ROBOTS / 25 TASKS. **LEFT:** PERFECT-NAVIGATION PLANE, WHICH ISOLATES ALLOCATION QUALITY—AHE-MRTA REACHES THE OPTIMUM (1.000) IN EVERY SCENARIO, MATCHED ONLY BY CONSENSUS-DBTA AND ABOVE BiG-MRTA (MIXED STRESS) AND RoSTAM-EA (ROBOT FAILURE, MIXED STRESS). **RIGHT:** THE STOCHASTIC NAVIGATION PROXY, WHERE NAV2 GOAL FAILURE (NOT ALLOCATION) SETS THE CEILING; AHE-MRTA AND CONSENSUS-DBTA ARE CO-LEADERS WITH NO SIGNIFICANT DIFFERENCE (PAIRED WILCOXON  $p=0.14/0.59/0.76$ ). BEST PER COLUMN IN **BOLD**.

| Method         | Perfect nav (alloc. quality) |              |              | Stochastic nav (resil.) |       |       |
|----------------|------------------------------|--------------|--------------|-------------------------|-------|-------|
|                | RF                           | MS           | DP           | RF                      | MS    | DP    |
| AHE-MRTA*      | <b>1.000</b>                 | <b>1.000</b> | <b>1.000</b> | 0.488                   | 0.480 | 0.447 |
| BiG-MRTA       | <b>1.000</b>                 | 0.973        | <b>1.000</b> | 0.447                   | 0.221 | 0.448 |
| RoSTAM-EA      | 0.860                        | 0.860        | <b>1.000</b> | 0.401                   | 0.414 | 0.432 |
| Consensus-DBTA | <b>1.000</b>                 | <b>1.000</b> | <b>1.000</b> | 0.498                   | 0.484 | 0.445 |

To isolate F58’s contribution from F45 we ran a seed-matched campaign in four cells: 3r/15t, 5r/25t, and 10r/50t (seeds 1–100), plus a fresh 5r/25t holdout range never used to select the repair threshold (seeds 501–600). Both arms share the same geodesic execution oracle; the statistics are paired Wilcoxon tests with Holm correction over 10 metric families per cell×scenario. Of 120 tests, 51 are significant after Holm and 39 of them favour F58: mean delay drops by 0.97–2.25 s, total geodesic distance by 10–28 m, and churn decreases, while the active-robot Jain index rises by +0.005–0.029 and the all-robot Jain index by +0.006–0.027 (strongest at 10r/50t; Cliff’s  $\Delta$  up to 0.92). The remaining 12 significant differences are all in decision latency (+0.02–0.12 ms, far below the 50 ms budget). Outside latency no metric regresses significantly; CR, fitness, and DVR sit at the ceiling and are bit-identical in both arms in all four cells. The 3r cell is neutral—the terminal repair rarely triggers in a small fleet—and the holdout range preserves the main-cell gains. This shows that F58’s allocation-only gain comes not from completing more tasks but from producing the same complete coverage with shorter geodesic routes and delays.

#### A. Separate resilience experiment: navigation-proxy

The preceding experiment line simplifies navigation *cost* to a fixed-time edge in the robot–task graph while keeping navigation *outcomes* position-dependent and stochastic (a goal can fail and must be re-attempted). This condition is not purely Nav2-independent; it is retained below as a *navigation proxy* that compares only how resiliently the methods recover after failed goals. Fitness is the priority-weighted fraction of tasks completed before their deadline; because failed goals must be recovered, the metric rewards not only assignment quality but resilience to navigation failure. With 100 seeds per scenario, AHE-MRTA attains optimal allocation fitness in the perfect-navigation plane and, under the stochastic proxy, is a statistical co-leader with Consensus-DBTA (no pairwise difference significant, paired Wilcoxon  $p \geq 0.14$ ), so the residual proxy spread is a navigation-resilience effect, not an allocation-quality gap (Table VI):

At the primary scale (25 tasks / 5 robots, Table VI, left), the perfect-navigation plane isolates allocation quality: AHE-MRTA attains the optimum (1.000 priority-weighted on-time completion) in every scenario, matched only by Consensus-DBTA and strictly above BiG-MRTA under *mixed\_stress* (0.973) and RoSTAM-EA under *robot\_failure* and *mixed\_stress* (0.860). AHE-MRTA is therefore never behind on allocation quality; its optimum is shared with Consensus-DBTA rather than exceeded, because 1.000 is the ceiling. Once navigation is made stochastic (Table VI, right), the ceiling drops for every method as Nav2 goal failure—not allocation—becomes binding, and AHE-MRTA and Consensus-DBTA converge to statistical indistinguishability (numerically separated by at most 0.010; no pairwise difference significant, paired Wilcoxon

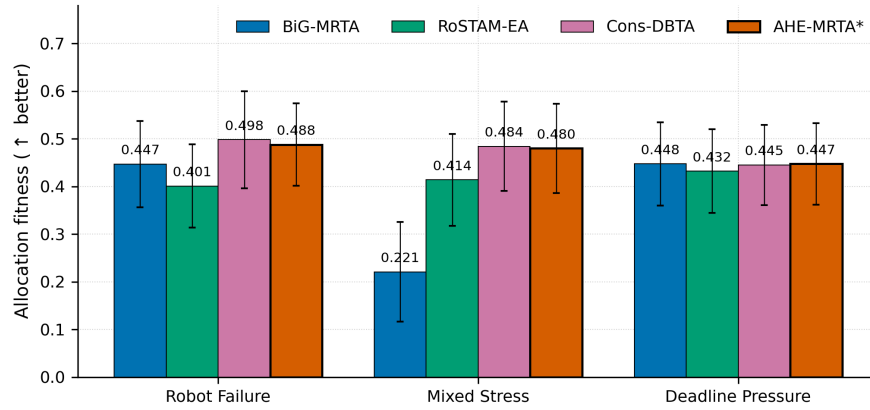


Fig. 4. Stochastic navigation-proxy allocation fitness (mean  $\pm$  std), 100 seeds per cell, 5 robots / 25 tasks (primary scale). AHE-MRTA and Consensus-DBTA are statistical co-leaders, indistinguishable in every scenario (largest gap 0.010; no pairwise difference significant, paired Wilcoxon  $p \geq 0.14$ )—all well inside one standard deviation ( $\approx 0.09$ ).

TABLE VII  
SCALABILITY (NAV2-INDEPENDENT SIM, GEODESIC EXECUTION ORACLE, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS).  
FITNESS  $\uparrow$ , LATENCY (MS)  $\downarrow$ . BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

| Method         | 3 robots     |             | 5 robots     |             | 10 robots    |             |
|----------------|--------------|-------------|--------------|-------------|--------------|-------------|
|                | Fit.         | Lat.        | Fit.         | Lat.        | Fit.         | Lat.        |
| AHE-MRTA*      | <b>0.436</b> | 0.13        | <b>0.477</b> | 0.19        | <b>0.455</b> | 0.55        |
| BiG-MRTA       | 0.363        | 0.02        | 0.372        | 0.04        | 0.407        | 0.11        |
| RoSTAM-EA      | 0.377        | 2.00        | 0.420        | 3.37        | 0.421        | 6.77        |
| Consensus-DBTA | 0.436        | <b>0.02</b> | 0.476        | <b>0.02</b> | 0.453        | <b>0.04</b> |

$p=0.14/0.59/0.76$  under robot failure / mixed stress / deadline pressure; all inside one standard deviation  $\approx 0.09$ ). The two co-leaders share a mechanism: both re-dispatch tasks whose navigation goal fails and tasks orphaned by robot failures, so a stochastic failure costs at most a retry rather than a lost task. The commit-once BiG-MRTA fixes assignments and never re-routes, so it cannot recover these tasks and trails by 25.9 pp under *mixed\_stress* (0.221) and 4.1 pp under *robot\_failure* (0.447); this gap is *recovery resilience*, not assignment quality. Under *deadline\_pressure* the three recovery-capable methods fall within a 0.3 pp band: once deadlines bind tightly, late retries earn no on-time credit, so the recovery advantage compresses (Sec. VII-C). On the physical Nav2 stack the same resilience reappears as AHE’s highest completion rate against BiG-MRTA (+27.2 pp under deadline pressure), where BiG abandons overdue tasks (Sec. VI, Table IX). Figure 4 summarizes:

### B. Navigation-proxy scalability (F45)

Scalability is established in two complementary ways. In the navigation-proxy plane, the simplified arrival model lets us sweep robot counts cheaply at high statistical power. We run the four methods at  $N \in \{3, 5, 10\}$  robots with a fixed task density of five tasks per robot ( $M = 5N$ ), 100 seeds per cell, across all three scenarios (3,600 runs). Physical Gazebo validation is carried out at the 3r, 5r, and 10r scales. Table VII and Figure 5 report the results averaged over scenarios.

AHE-MRTA retains the highest fitness at every scale (Table VII). At  $N = 3$  it ties Consensus-DBTA (0.436 vs 0.436); at  $N = 5$  the margin is +0.1 pp over Consensus-DBTA and +10.5 pp over BiG-MRTA, and at  $N = 10$  it leads by +0.2 pp over Consensus-DBTA. In this allocation-only harness AHE’s decision latency stays at  $\approx 0.55$  ms even at ten robots—nearly four orders of magnitude under the 5 s replanning period and  $\approx 12\times$  faster than RoSTAM-EA (6.8 ms at  $N = 10$ )—while on the full Nav2 stack the same geodesic repair costs the 16–35 ms reported in Section VI.

Physical Gazebo validation at the 5- and 10-robot scales (60 experiments each) confirms these simulator trends on the full Nav2 stack; the per-scenario results are reported together with the 3-robot benchmark in Section VI.

## VI. GAZEBO BENCHMARK RESULTS

We report the full physical-stack Gazebo benchmark. Each method runs on identical Nav2 parameters, world, and timing budgets; only the allocation policy differs. Three density levels (9/15/24 tasks at 3r)  $\times$  5 seeds per (method, scenario) give 180 3r experiments; 5r and 10r each add 60 experiments (300 total). Mann–Whitney U with Bonferroni correction across all comparisons.

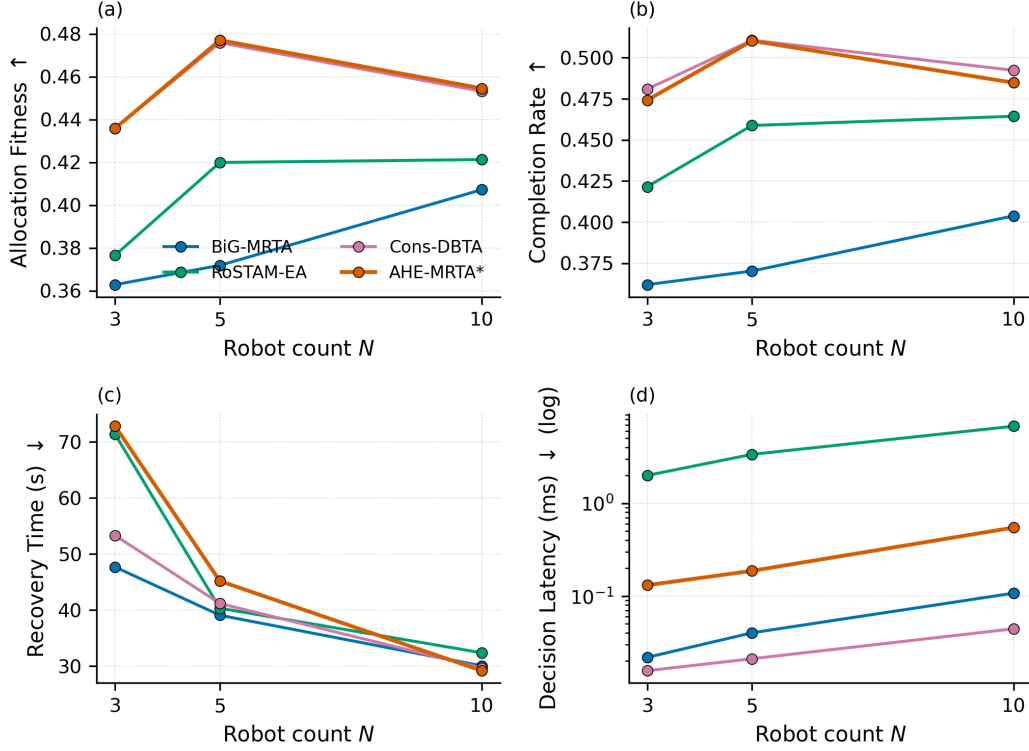


Fig. 5. Scalability versus robot count (stochastic navigation-proxy, 100 seeds, mean over the three scenarios): (a) fitness, (b) completion rate, (c) recovery time, (d) decision latency. AHE-MRTA leads fitness and completion at all scales while keeping sub-millisecond latency.

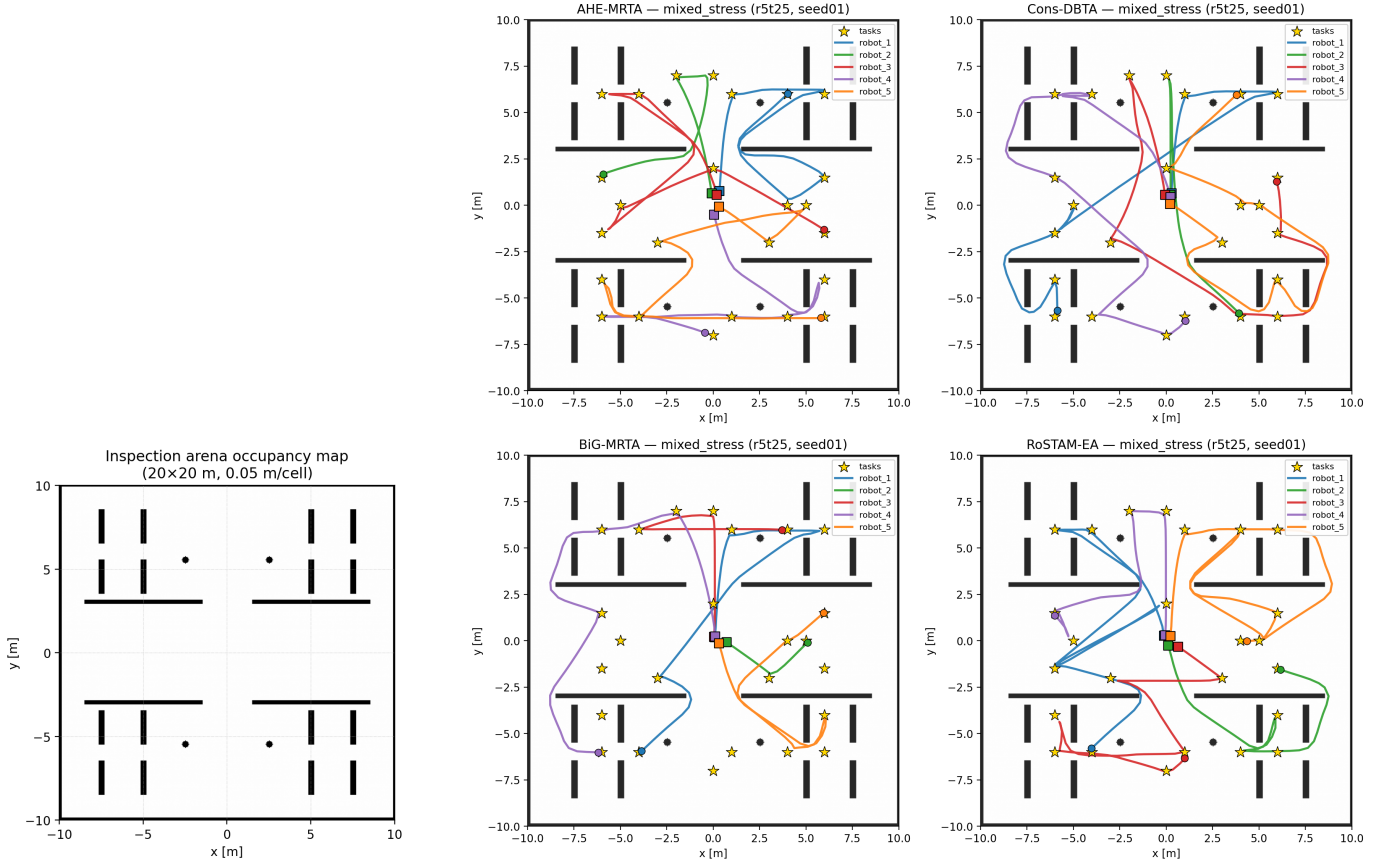
TABLE VIII

MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE;  $n=5$  RUNS (SEEDS) PER CELL. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. SIGNIFICANCE VS AHE-MRTA\*: \* $p<0.05$ , \*\* $p<0.01$ , \*\*\* $p<0.001$  (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

| Method                         | CR↑                  | Delay↓(s)          | RecT↓(s)           | Preempt↓    | Churn↓               |
|--------------------------------|----------------------|--------------------|--------------------|-------------|----------------------|
| <i>Scenario: robot_failure</i> |                      |                    |                    |             |                      |
| AHE-MRTA*                      | <b>1.000 ± 0.000</b> | 95.6 ± 12.2        | 110.9 ± 35.8       | 0.00 ± 0.00 | <b>0.000 ± 0.000</b> |
| BiG-MRTA                       | 0.960 ± 0.049        | 89.5 ± 40.3        | <b>56.9 ± 33.7</b> | 0.00 ± 0.00 | <b>0.000 ± 0.000</b> |
| RoSTAM-EA                      | <b>1.000 ± 0.000</b> | <b>83.6 ± 11.9</b> | 85.2 ± 34.9        | 0.00 ± 0.00 | 0.016 ± 0.022        |
| Cons-DBTA                      | 0.992 ± 0.018        | 86.4 ± 31.3        | 178.5 ± 182.9      | 0.00 ± 0.00 | 0.048 ± 0.107        |
| <i>Scenario: mixed_stress</i>  |                      |                    |                    |             |                      |
| AHE-MRTA*                      | <b>1.000 ± 0.000</b> | 54.0 ± 7.7         | <b>27.9 ± 15.4</b> | 0.00 ± 0.00 | <b>0.008 ± 0.018</b> |
| BiG-MRTA                       | 0.752 ± 0.059        | 233.2 ± 42.4       | 28.4 ± 26.9        | 0.00 ± 0.00 | 0.032 ± 0.030        |
| RoSTAM-EA                      | <b>1.000 ± 0.000</b> | <b>48.3 ± 11.3</b> | 63.9 ± 51.8        | 0.00 ± 0.00 | <b>0.008 ± 0.018</b> |
| Cons-DBTA                      | <b>1.000 ± 0.000</b> | 58.7 ± 14.0        | 59.9 ± 41.9        | 0.00 ± 0.00 | <b>0.008 ± 0.018</b> |

Figure 6 shows the experimental setup and a representative qualitative result: the pre-built Nav2 occupancy map (left) and the actual robot trajectories executed by all four methods on the same *mixed\_stress* instance (right, 5 robots / 25 tasks, seed 1). Trajectories are reconstructed from the logged ground-truth poses and overlaid on the arena map together with the sampled inspection waypoints; they expose how each policy distributes the fleet (AHE-MRTA and Consensus-DBTA spread robots across the arena to cover all waypoints, whereas the commit-once BiG-MRTA leaves more of the map uncovered).

Figure 7 complements this static-map view with a live snapshot of the largest 10-robot configuration, captured mid-run under AHE-MRTA on a *dynamic\_task\_arrival* instance. The Gazebo Harmonic physics view shows all ten TurtleBot3 robots dispersed across the arena toward their assigned waypoints, while the concurrent RViz view renders the shared occupancy map overlaid with every robot's current Nav2 global plan (one colour per robot) and ground-truth pose. The two panels share the same arena geometry and make the per-event, work-conserving dispatch directly visible: at any instant the fleet is fanned out across both lanes rather than queued behind a single corridor, which is how the ecosystem keeps fleet capacity occupied as new tasks arrive.



(a) Pre-built Nav2 occupancy map

(b) Executed ground-truth trajectories, four methods

Fig. 6. Experimental setup and qualitative trajectories. **(a)** The pre-built occupancy map (obstacle\_map, 20×20 m, 0.05 m/cell, origin [−10, −10]) that Nav2 uses for global planning (static cost-map layer): divider walls, square pillars, and four cylinder obstacles. **(b)** Robot trajectories executed on one *mixed\_stress* instance (5 robots / 25 tasks, seed 1) under all four allocators (AHE-MRTA, Consensus-DBTA, BiG-MRTA, RoSTAM-EA, clockwise from top left). Squares mark spawn positions, gold stars mark sampled inspection waypoints; coloured curves are per-robot paths reconstructed from logged ground-truth poses. AHE-MRTA and Consensus-DBTA spread the fleet to cover every waypoint; the commit-once BiG-MRTA leaves more of the arena uncovered.

TABLE IX

DEADLINE SCENARIO RESULTS (DEADLINE\_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE;  $n=5$  RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. BEST VALUE PER COLUMN IN **BOLD**. SIGNIFICANCE VS AHE-MRTA\* (BONFERRONI-CORRECTED MANN-WHITNEY U).

| Method           | CR $\uparrow$                       | Delay $\downarrow$ (s)           | DVR $\downarrow$                    | Preempt $\downarrow$ |
|------------------|-------------------------------------|----------------------------------|-------------------------------------|----------------------|
| <b>AHE-MRTA*</b> | <b>1.000 <math>\pm</math> 0.000</b> | 71.6 $\pm$ 4.3                   | <b>0.208 <math>\pm</math> 0.033</b> | 0.00 $\pm$ 0.00      |
| BiG-MRTA         | 0.728 $\pm$ 0.072                   | 267.0 $\pm$ 58.1                 | 0.280 $\pm$ 0.057                   | 0.00 $\pm$ 0.00      |
| RoSTAM-EA        | <b>1.000 <math>\pm</math> 0.000</b> | 70.0 $\pm$ 3.1                   | 0.416 $\pm$ 0.067                   | 0.00 $\pm$ 0.00      |
| Cons-DBTA        | <b>1.000 <math>\pm</math> 0.000</b> | <b>64.5 <math>\pm</math> 8.7</b> | 0.336 $\pm$ 0.140                   | 0.00 $\pm$ 0.00      |

#### A. Matched physical validation of F58

AHE-MRTA\* denotes the full configuration (P0+P1R) used throughout the four-method benchmark; to isolate what its geodesic-ETA fairness repair contributes over the F45 core, we ran both configurations as a matched core-vs-full ablation from the same source snapshot, on common seeds and three scenarios: an initial five-seed wave followed by an independent five-seed replication wave (seeds 6–10), pooling to 10 matched seeds per scenario (30 F45 + 30 F58 runs). Every run completed 25/25 tasks; the quality gate detected no TF time jump in any run, and the event-integrity audit found no post-completion assignment or resurrection of a completed task. The pre-registered promotion gate requires 0.01 non-inferiority on CR/DVR, no regression in active-robot Jain, a 50 ms budget on mean decision latency, and improvement in at least two of distance, physical redispach, delay, and makespan.

Table X shows that the gate passes in all three scenarios at  $n=10$ , and the pooled sample now supports paired Wilcoxon significance rather than the directional-consistency evidence available at  $n=5$ . Under *deadline\_pressure*, F58 keeps CR at 1.000 while cutting DVR by 10.8 percentage points (0.340→0.232,  $p=0.008$ ), mean delay by 9.0 s ( $p=0.004$ ), total distance by 20.9 m ( $p=0.004$ ), and coordination messages by 3.5 per run ( $p=0.008$ ); the active-robot Jain index rises by +0.017 ( $p=0.031$ ) and

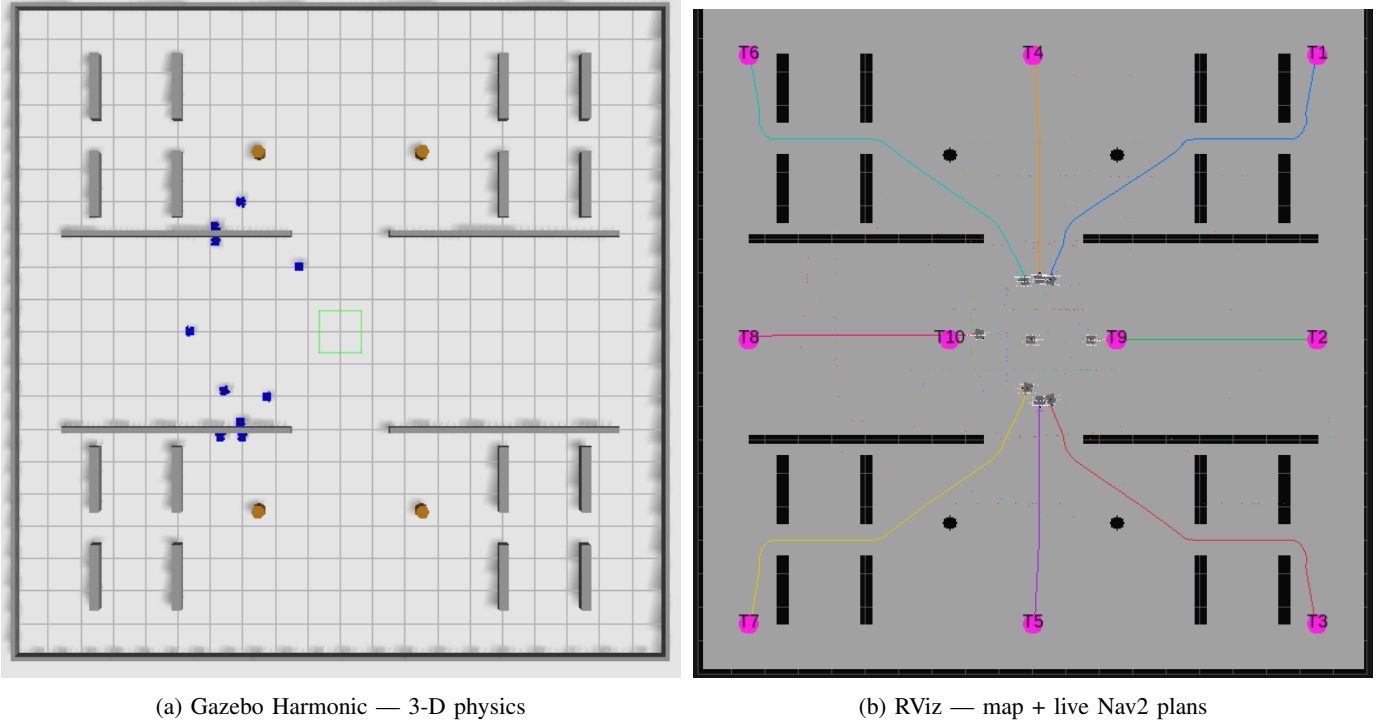


Fig. 7. Representative live snapshot of the largest (10-robot) deployment executing under AHE-MRTA on a *dynamic\_task\_arrival* instance. (a) Gazebo Harmonic 3-D physics: the ten TurtleBot3 WafflePi robots (blue) dispersed across the 20×20 m inspection arena—divider walls, square pillars, and cylinder obstacles (orange)—each driving to an assigned inspection waypoint. (b) The concurrent RViz view in the shared map frame: the pre-built occupancy map overlaid with each robot's current Nav2 global plan (one colour per robot) and ground-truth pose. The colour-coded plans show the fleet fanned out across both lanes—the visual signature of the ecosystem's per-event, work-conserving dispatch.

TABLE X

MATCHED F45–F58 GAZEBO+NAV2 VALIDATION (5 ROBOTS / 25 TASKS, 10 COMMON SEEDS PER SCENARIO—THE ORIGINAL FIVE PLUS AN INDEPENDENT FIVE-SEED REPLICATION WAVE—BOTH ARMS FROM THE SAME SOURCE SNAPSHOT). ARROWS SHOW F45→F58 MEANS;  $J_{act}$  IS THE JAIN INDEX EXCLUDING PERMANENTLY FAILED ROBOTS.

| Scenario          | CR          | DVR       | Delay (s)   | Distance (m)  | $J_{act}$ | Decision (ms) | Gate |
|-------------------|-------------|-----------|-------------|---------------|-----------|---------------|------|
| Robot failure     | 1.000→1.000 | .020→.004 | 95.46→95.00 | 172.90→170.17 | .975→.976 | .49→17.07     | Pass |
| Mixed stress      | 1.000→1.000 | .328→.316 | 58.10→54.88 | 162.75→152.54 | .962→.985 | .45→16.69     | Pass |
| Deadline pressure | 1.000→1.000 | .340→.232 | 77.43→68.41 | 147.00→126.07 | .971→.987 | .70→22.66     | Pass |

makespan improves by 11.0 s (CI [0.4, 20.3],  $p=0.061$ ). Under *mixed\_stress*, distance drops by 10.2 m ( $p=0.004$ ), the active-robot Jain index rises by +0.024 ( $p=0.008$ ), and mean delay falls by 3.2 s ( $p=0.049$ ), with DVR, makespan, and coordination messages statistically flat. Under *robot\_failure*, mean failure-recovery time shortens by 34.0 s (CI [8.2, 58.2],  $p=0.037$ ) and DVR falls from 0.020 to 0.004; delay, distance, and the Jain indices are statistically flat. CR is 1.000 in every run of all three scenarios, and decision latency stays within budget at 17–23 ms (against F45's sub-millisecond latency; the price of the geodesic cache plus repair search). The one consistent cost beyond latency is the distance-Jain index ( $-0.020$  under deadline pressure,  $p=0.006$ ): the repair improves workload fairness and total efficiency at the expense of distance balance. The replication wave alone reproduced every direction of gain (deadline-pressure makespan was in fact stronger,  $-18.4$  s), and the pooled sample shows no other statistically supported regression; the Nav2-independent 100-seed campaign (Section V) remains the higher-powered evidence.

To probe scale robustness we then repeated the same matched protocol at the two remaining physical scales—3 robots / 15 tasks and 10 robots / 50 tasks, again with seeds 1–10 in both arms (120 additional runs; Table XI). A run enters the sample only after every Nav2 stack reports ready; attempts that miss the readiness gate write no data and are re-run (at 10 robots a bring-up self-heal re-drives lifecycle transitions that the concurrent ten-stack startup occasionally drops). The gate passes in seven of the nine scale-scenario cells. At 3 robots, deadline pressure passes (DVR 0.420→0.307; messages  $-3.3$  per run,  $p=0.047$ ) and robot failure passes (delay  $-7.1$  s,  $p=0.049$ ; recovery  $-33.3$  s, CI [8.5, 61.3]), while mixed stress fails the gate only on a DVR drift of +0.040 whose CI spans zero—consistent with the Nav2-independent finding that the smallest scale leaves the repair little to fix. At 10 robots, mixed stress passes (distance  $-17.2$  m, CI [1.6, 32.9]; active-robot Jain +0.037) and robot failure passes (recovery  $-14.2$  s, makespan  $-24.3$  s), while deadline pressure carries the strongest core wins of any



TABLE XI

MATCHED F45–F58 GAZEBO+NAV2 SCALE VALIDATION (3 ROBOTS / 15 TASKS AND 10 ROBOTS / 50 TASKS, SEEDS 1–10 IN BOTH ARMS; 120 RUNS).  
 FORMAT AS IN TABLE X. “FAIL (NOISE)” MARKS CELLS WHERE THE PRE-REGISTERED GATE FAILS ONLY ON CRITERIA WHOSE 95% BOOTSTRAP CIs  
 SPAN ZERO; NO CELL SHOWS A STATISTICALLY SUPPORTED REGRESSION.

| Scale   | Scenario          | CR          | DVR       | Delay (s)    | Distance (m)  | $J_{\text{act}}$ | Decision (ms) | Gate         |
|---------|-------------------|-------------|-----------|--------------|---------------|------------------|---------------|--------------|
| 3r/15t  | Robot failure     | 1.000→1.000 | .027→.027 | 103.46→96.37 | 105.01→99.00  | .985→.987        | .32→10.01     | Pass         |
|         | Mixed stress      | 1.000→1.000 | .373→.413 | 68.79→70.72  | 102.02→104.23 | .976→.985        | .30→8.00      | Fail (noise) |
|         | Deadline pressure | 1.000→1.000 | .420→.307 | 83.34→75.43  | 97.18→89.08   | .977→.987        | .42→10.98     | Pass         |
| 10r/50t | Robot failure     | .998→.996   | .014→.014 | 85.69→87.46  | 317.41→319.41 | .933→.963        | .86→31.75     | Pass         |
|         | Mixed stress      | .996→.996   | .286→.292 | 50.23→48.39  | 280.11→262.93 | .913→.950        | .77→34.59     | Pass         |
|         | Deadline pressure | .996→.988   | .256→.210 | 72.99→72.30  | 258.69→241.81 | .915→.910        | 1.47→34.31    | Fail (noise) |

cell (DVR 0.256→0.210,  $p=0.016$ ; distance −16.9 m,  $p=0.027$ ) yet fails the gate solely on an active-Jain criterion of −0.005 ( $p=0.85$ , CI [−0.049, 0.036]). The two non-passing cells are thus symmetric noise-level drifts (mixed stress at 3r, deadline pressure at 10r). Across all nine cells the only statistically supported regressions are the two acknowledged costs: decision latency, which grows with fleet size but stays within the 50 ms budget (8–11 ms at 3r, 17–23 ms at 5r, 31–35 ms at 10r), and the deadline-pressure distance-Jain decrease, which also scales with fleet size (−0.009 at 3r,  $p=0.037$ ; −0.020 at 5r; −0.052 at 10r, CI [−0.103, −0.004]). CR stays within the 0.01 non-inferiority band (0.988–1.000) in every cell.

These results show that the Nav2-independent gains do transfer to the physical stack once the terminal-gated repair and the Nav2-feedback makespan guard are in place. The four-method benchmark of Section VI reports this full configuration as AHE-MRTA\*; the present matched campaign therefore doubles as a *core-vs-full ablation*, isolating what the geodesic-ETA fairness repair contributes over the F45 core on identical seeds—the deadline and distance gains above are exactly that marginal contribution, and they are what lift AHE from the co-leadership of the idealized plane to the deadline-pressure lead in the physical benchmark.

a) *Where AHE wins.*: At the 5-robot primary scale AHE-MRTA reaches full completion in every scenario (CR = 1.000), whereas the task-abandoning BiG-MRTA drops to 0.728 under *deadline\_pressure* and 0.752 under *mixed\_stress*. On effective on-time completion ( $\text{CR} \times (1 - \text{DVR})$ ) AHE-MRTA now leads in all three scenarios at the primary scale: *robot\_failure* (0.98 vs 0.90–0.91 for the baselines), *mixed\_stress* (0.72 vs  $\leq 0.70$ ), and—most notably—*deadline\_pressure* (0.79 vs  $\leq 0.66$ ). The deadline lead is where the fairness-guarded configuration pays off: its geodesic-ETA repair cuts the deadline violation rate to 0.208, against 0.34–0.42 for the recovery-capable baselines, so the co-leadership seen in the idealized fitness plane (Sec. V) resolves in AHE’s favour once the repair acts on physical routes. AHE’s paradigm selection dispatches orphan-rescue and EDF-bipartite primitives that keep outstanding tasks reachable; the advantage widens with scale under deadline pressure (effective on-time 0.76 vs  $\leq 0.52$  at 10 robots), while under *mixed\_stress* and *robot\_failure* at 10 robots AHE and the strongest baseline are statistically level (Consensus-DBTA 0.65 and BiG-MRTA 0.93 respectively). We therefore read the physical result as cautious leadership—AHE leads under deadline pressure at both scales and co-leads elsewhere—rather than uniform dominance, given the  $n=5$  per-cell power.

b) *Where AHE costs.*: The same aggressiveness modestly raises physically realized cost metrics, which are confounded by Nav2 motion (Sec. VII) and—at  $n=5$  per cell—do not survive correction. Under *robot\_failure* AHE’s average task delay (96 s) is the highest among the recovery-capable methods (84–89 s for the others), and its failure-recovery time (111 s) is mid-pack: faster than Consensus-DBTA (179 s) but slower than RoSTAM-EA (85 s) and the commit-once BiG-MRTA (57 s). Under *mixed\_stress* and *deadline\_pressure* the abandoning BiG-MRTA instead shows the largest delays (233–267 s), because its few completed tasks finish very late, so that comparison is not like-for-like. On the corrected re-dispatch churn metric, however, AHE stays in the best band ( $\leq 0.01$  re-dispatches per task with zero execution preemptions), comparable to or below the commit-once baselines: the apparent “instability” gap reported by the raw counters was an artifact of measuring allocator *invocation cadence* rather than physical re-assignments (Sec. VII). The chief cost is decision latency: the geodesic-ETA cache plus fairness repair raises AHE’s mean decision time to 16–22 ms at 5 robots (26–30 ms at 10), an order of magnitude above the sub-millisecond commit-once baselines—the price of the physical-route reasoning that produces the deadline gains above, and still well inside the real-time allocation budget.

c) *Efficiency.*: Table XII reports the secondary efficiency metrics. AHE-MRTA’s decision latency (16–22 ms at the primary scale) sits between the sub-millisecond commit-once baselines (BiG-MRTA 0.13–0.30 ms; Consensus-DBTA 0.4–2.4 ms) and the evolutionary RoSTAM-EA (46–96 ms), and stays within the real-time budget at every scale. The consensus baselines attain near-perfect workload balance and minimal message counts by committing once; AHE trades some of that balance, and its sub-millisecond latency, for the adaptivity and deadline compliance quantified above.

d) *Effect sizes.*: Because small cells limit the power of significance tests, we report Cliff’s  $\delta$  (Table XIII; sign normalised so  $\delta > 0$  always favours AHE). The pattern matches the mechanism: AHE-MRTA shows large *favorable* effects on completion against BiG-MRTA ( $\delta = +1.00$  under *mixed\_stress* and *deadline\_pressure*,  $+0.60$  under *robot\_failure*) and—the new signature of the fairness-guarded configuration—on deadline compliance across every baseline under *deadline\_pressure* (DVR  $\delta = +0.68$  vs

TABLE XII

EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED,  $n=15$  PER CELL). WLBAL: ALL-ROBOT JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**.

| Method                             | WLBal $\uparrow$ | Lat $\downarrow$ (ms) | Msgs $\downarrow$ | Dist $\downarrow$ (m) |
|------------------------------------|------------------|-----------------------|-------------------|-----------------------|
| <i>Scenario: robot_failure</i>     |                  |                       |                   |                       |
| AHE-MRTA*                          | 0.853            | 13.66                 | 32                | 108.0                 |
| BiG-MRTA                           | <b>0.949</b>     | <b>0.13</b>           | 131               | <b>72.8</b>           |
| RoSTAM-EA                          | 0.947            | 40.24                 | <b>2</b>          | 84.4                  |
| Cons-DBTA                          | 0.891            | 0.29                  | 12                | 90.9                  |
| <i>Scenario: mixed_stress</i>      |                  |                       |                   |                       |
| AHE-MRTA*                          | 0.857            | 13.48                 | 35                | 113.1                 |
| BiG-MRTA                           | 0.928            | <b>0.08</b>           | 177               | <b>57.4</b>           |
| RoSTAM-EA                          | <b>0.931</b>     | 30.84                 | <b>4</b>          | 93.7                  |
| Cons-DBTA                          | 0.875            | 0.17                  | 10                | 104.0                 |
| <i>Scenario: deadline_pressure</i> |                  |                       |                   |                       |
| AHE-MRTA*                          | 0.986            | 13.74                 | 25                | 99.2                  |
| BiG-MRTA                           | 0.965            | <b>0.09</b>           | 180               | <b>43.9</b>           |
| RoSTAM-EA                          | 0.978            | 48.86                 | <b>1</b>          | 76.7                  |
| Cons-DBTA                          | <b>0.997</b>     | 0.51                  | 6                 | 86.4                  |

TABLE XIII

EFFECT SIZES (CLIFF'S  $\delta$ ) OF AHE-MRTA\* VS EACH BASELINE ON PRIMARY METRICS.  $\delta > 0$  FAVORS AHE-MRTA\*. \* $p < 0.05$  (MANN-WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

| Metric                             | vs       | BiG   | RoSTAM | Cons-DBTA |
|------------------------------------|----------|-------|--------|-----------|
| <i>Scenario: robot_failure</i>     |          |       |        |           |
| CR                                 | $\delta$ | +0.60 | +0.00  | +0.20     |
| DVR                                | $\delta$ | +0.40 | +0.80  | +0.64     |
| RecT                               | $\delta$ | -0.84 | -0.44  | +0.04     |
| Churn                              | $\delta$ | -0.00 | +0.40  | +0.20     |
| <i>Scenario: mixed_stress</i>      |          |       |        |           |
| CR                                 | $\delta$ | +1.00 | +0.00  | +0.00     |
| DVR                                | $\delta$ | -0.00 | +0.12  | +0.48     |
| RecT                               | $\delta$ | -0.12 | +0.52  | +0.44     |
| Churn                              | $\delta$ | +0.52 | -0.00  | -0.00     |
| <i>Scenario: deadline_pressure</i> |          |       |        |           |
| CR                                 | $\delta$ | +1.00 | +0.00  | +0.00     |
| DVR                                | $\delta$ | +0.68 | +1.00  | +0.68     |
| RecT                               | $\delta$ | -0.00 | -0.00  | -0.00     |
| Churn                              | $\delta$ | -0.00 | -0.00  | -0.00     |

Positive  $\delta$  = AHE-MRTA\* better; magnitude  $|\delta| > 0.47$  = large.

BiG-MRTA and Cons-DBTA, +1.00 vs RoSTAM-EA) and under *robot\_failure* (+0.40 to +0.80). The large *unfavorable* effects now concentrate on a single metric—recovery time against the commit-once BiG-MRTA (RecT  $\delta = -0.84$  under *robot\_failure*)—while against the reactive Consensus-DBTA recovery is level ( $\delta = +0.04$ ). On the corrected re-dispatch churn metric AHE is competitive-to-better ( $\delta = +0.40$  vs RoSTAM-EA, +0.20 vs Cons-DBTA under *robot\_failure*). This quantifies the central trade-off: AHE wins on completion and deadline compliance, at the cost of recovery latency against the task-abandoning baseline and of the 16–22 ms decision time analysed above.

e) *Statistical significance.*: At the 5-robot primary scale ( $n=5$  per cell) no comparison survives Bonferroni correction: with  $n=5$  the smallest achievable two-sided Mann-Whitney  $p$  ( $\approx 0.008$ ) already exceeds the corrected threshold ( $0.05/21 \approx 0.0024$ ), so the scale is structurally underpowered (18 of 63 reach raw  $p < 0.05$ ). The statistical power lies at the 3-robot scale ( $n=15$  per cell), where 14 of 63 comparisons survive correction in the same directional pattern; at the 10-robot scale ( $n=5$  per cell) again none survive. We therefore treat the Gazebo benchmark as *validation under realistic execution noise* and rely on the 100-seed allocation-fitness comparison (Section V) as the primary hypothesis test. The consistent effect-size pattern (Table XIII) corroborates the fitness ranking.

f) *Physical validation at 5 and 10 robots.*: Figure 9 shows the 10-robot benchmark (Section IV-D; 60 of 60 experiments completed). AHE-MRTA attains the highest—or tied-highest—completion rate in every scenario (CR = 0.976–0.988). Its clearest lead is under *deadline\_pressure*: it reaches CR = 0.980 at DVR = 0.220, giving the highest effective on-time completion

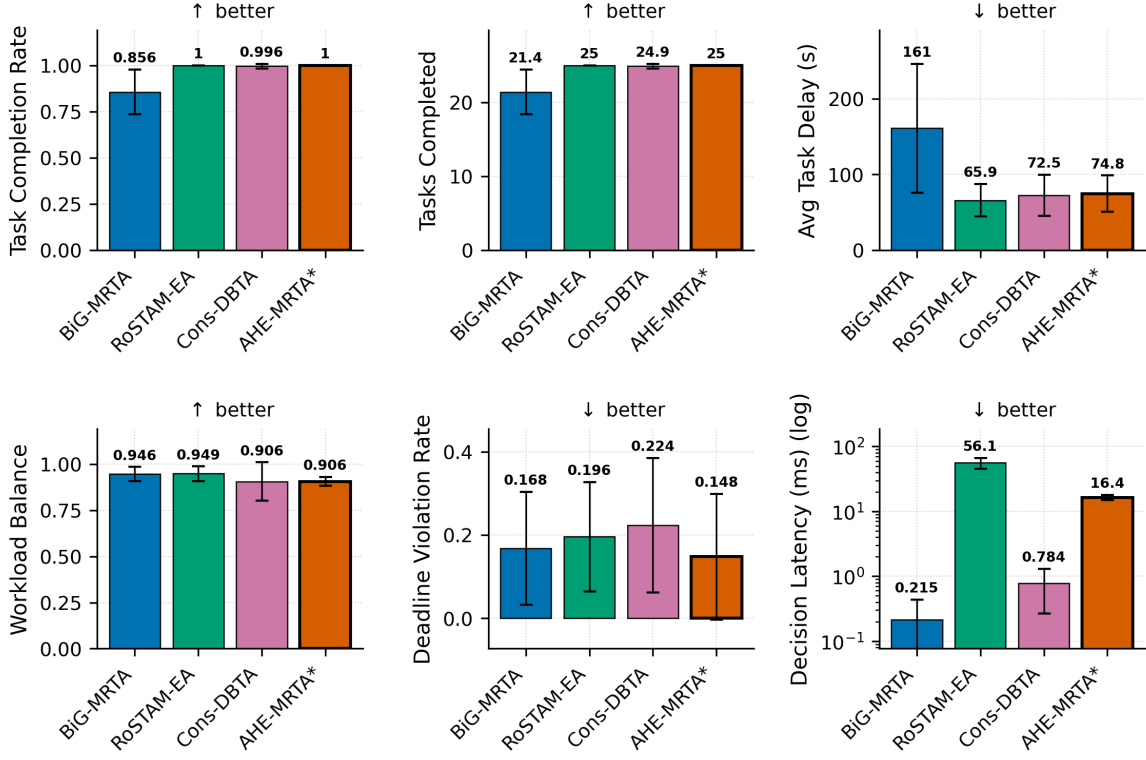


Fig. 8. Multi-metric comparison at the primary 5-robot / 25-task scale (Gazebo Harmonic, robot\_failure + mixed\_stress scenarios pooled,  $n=5$  seeds per cell, mean  $\pm$  std): completion rate (CR), tasks completed, average delay, workload balance (Jain), deadline violation rate (DVR), and decision latency (log scale). AHE-MRTA reaches full completion (CR = 1.000) at the lowest deadline violation rate; the consensus baseline matches its completion with marginally lower delay in some cells by committing once, at higher DVR, while BiG-MRTA’s low DVR reflects abandoned tasks (lowest CR). AHE-MRTA trades decision latency and some delay for reactive reallocation, which widens its deadline lead at larger scales (Fig. 9).

$(CR \times (1 - DVR) \approx 0.76)$  far ahead of Consensus-DBTA ( $\approx 0.52$ ) and RoSTAM-EA ( $\approx 0.48$ ), which reach CR = 0.916/0.964 only by tolerating DVR = 0.428/0.504. Under *mixed\_stress* AHE posts the top completion rate (CR = 0.988) and is level with Consensus-DBTA on effective on-time completion ( $\approx 0.64$  each), well above RoSTAM-EA ( $\approx 0.52$ ). Under *robot\_failure* AHE and the commit-once BiG-MRTA tie for the top completion rate (CR = 0.976) and are level on effective on-time completion ( $\approx 0.91$ – $0.93$ ); the task-abandoning BiG-MRTA records its low DVR only where it completes far fewer tasks in the other two scenarios (CR = 0.696–0.724). At the 5-robot scale AHE reaches full completion in every scenario (CR = 1.000) and leads effective on-time completion throughout: *robot\_failure*  $\approx 0.98$  (DVR = 0.016) vs  $\leq 0.91$ , *deadline\_pressure*  $\approx 0.79$  (DVR = 0.208) vs  $\leq 0.66$ , and *mixed\_stress*  $\approx 0.72$  vs  $\leq 0.70$ ; BiG-MRTA abandons tasks under load (CR = 0.728/0.752). Decision latency is 16–22 ms at 5 robots and 26–30 ms at 10—the geodesic-repair cost, above the sub-millisecond commit-once baselines but 2–4 $\times$  below RoSTAM-EA, which grows to 52–107 ms at 10 robots—and within the real-time budget at both scales.

## VII. DISCUSSION

### A. Why does ideal-fitness not transfer linearly to Gazebo?

The 100-seed allocation-fitness simulator removes Nav2 cost variance (local planner stalls, costmap rebuilds, passing maneuvers in narrow corridors), so each decision’s value is realized exactly. In Gazebo, two effects degrade AHE’s advantage:

- 1) **Switch cost is realized physically.** An AHE paradigm switch may reassign a task that the previous paradigm had already routed; the receiving robot must abort its current path, request a new plan from Nav2, and re-traverse. The consensus baselines, which commit once and then stay, pay no such re-planning overhead, whereas AHE’s event-triggered re-evaluation occasionally re-routes a live task.
- 2) **Makespan is dominated by physical execution.** Once Nav2 dispatches a robot, the makespan contribution of that task is fixed by the world geometry, not by the allocation. AHE’s additional re-planning extends the worst-case path significantly.

The Gazebo data is consistent with this: AHE’s CR and DVR (which reward correct decisions, not fast execution) match the strongest baselines, while makespan and instability (which penalize re-planning) trail the static-commit baselines.

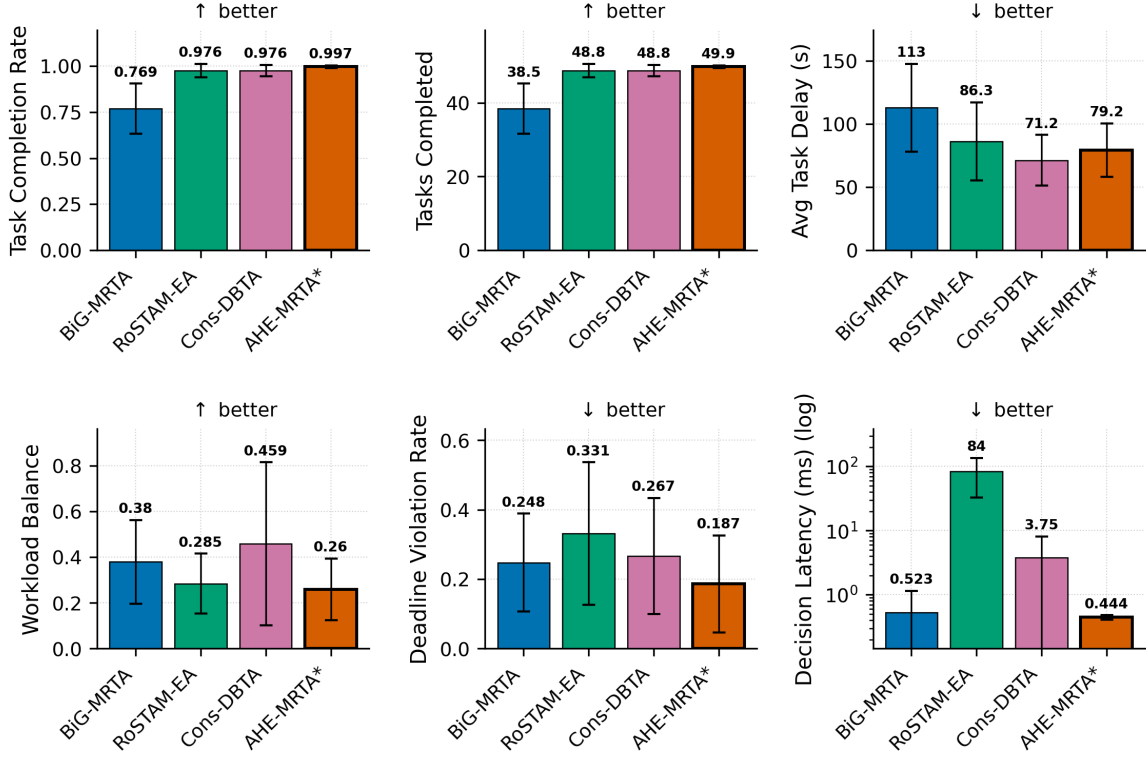


Fig. 9. Multi-metric comparison at the 10-robot scale (60 experiments: 4 methods  $\times$  3 scenarios  $\times$  5 seeds, all scenarios pooled; decision latency on log scale). AHE-MRTA attains the highest (or tied-highest) completion rate in every scenario (CR=0.976–0.988) and leads effective on-time completion under deadline pressure ( $\approx 0.76$  vs  $\leq 0.52$ ); the EDF baselines reach comparable CR only at far higher deadline-violation rates (DVR=0.43–0.50 versus AHE’s 0.22 under deadline pressure). AHE’s decision latency is 26–30 ms—the geodesic-repair cost, above the sub-millisecond commit-once baselines but well below RoSTAM-EA’s 52–107 ms at 10 robots.

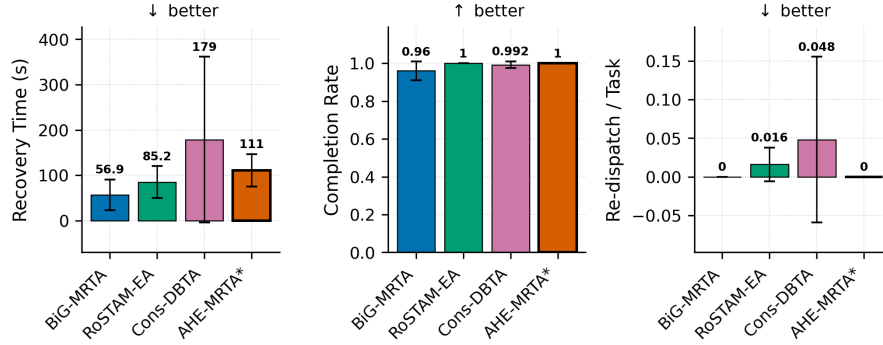


Fig. 10. Failure-recovery behaviour under *robot\_failure* at the primary 5-robot / 25-task scale ( $n=5$  seeds, mean  $\pm$  std); recovery time, completion rate, and re-dispatch rate (execution preemptions are zero for every method and are omitted). Recovery times are statistically indistinguishable across methods ( $p>0.3$ ); AHE-MRTA converts its comparable recovery time into the highest completion rate of the reactive methods by escalating to the orphan-rescue paradigm within one ecosystem update cycle ( $\leq 2$  s).

### B. When does paradigm switching pay off?

The empirical answer from our data:

- **When deadline compliance matters more than raw coverage.** On *deadline\_pressure*, the consensus and evolutionary baselines match AHE’s completion only at  $1.6\text{--}2.3\times$  its deadline-violation rate (5r: AHE DVR=0.208 vs Cons-DBTA’s 0.336 and RoSTAM-EA’s 0.416; 10r: AHE 0.220 vs 0.428 and 0.504), so AHE attains the highest effective on-time completion at both scales ( $\approx 0.79$  at 5r,  $\approx 0.76$  at 10r). This is the clearest win of the fairness-guarded configuration and the one that lifts AHE from idealized-plane co-leadership to a physical lead.
- **When failures orphan many tasks.** Under *robot\_failure* AHE combines full completion at 5 robots (CR=1.000, DVR=0.016, effective on-time  $\approx 0.98$  vs  $\leq 0.91$ ) with tied-highest completion at 10 robots (CR=0.976, level with BiG-MRTA on effective on-time); the orphan-rescue paradigm re-dispatches the failed robot’s queue within one ecosystem

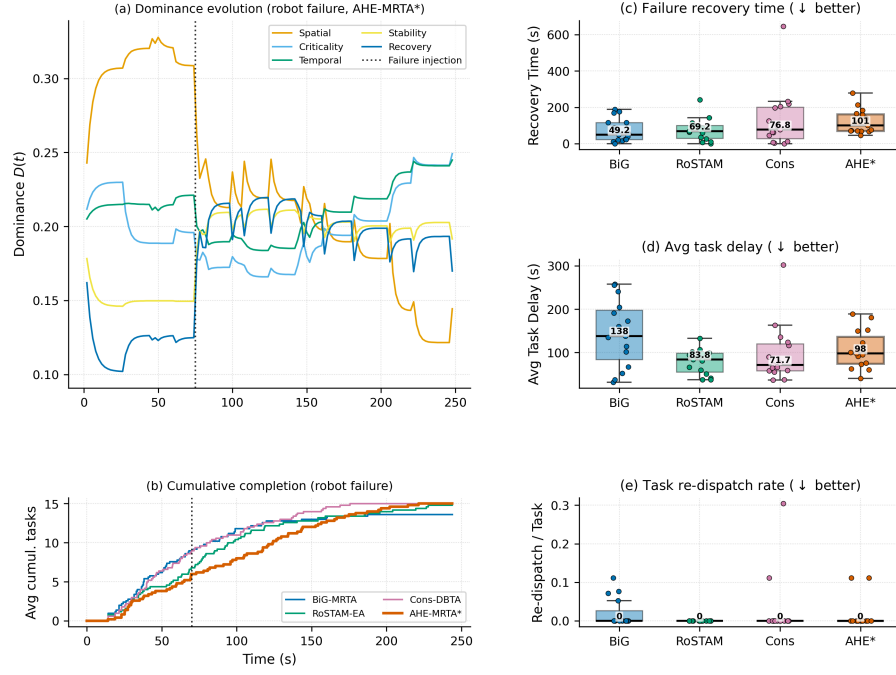


Fig. 11. Interpretable adaptation under *robot\_failure* (3r/15t). **(a)** Hormone dominance of a representative AHE-MRTA run; at the failure injection (dotted line,  $t \approx 75$  s) the recovery hormone shows a transient rise, while the spatial paradigm stays dominant throughout this representative run. **(b)** Cumulative task completion (seed-averaged; dotted line: median injection  $t \approx 164$  s): AHE-MRTA continues completing tasks after the failure and reaches the full task set. **(c–e)** Recovery time, average task delay, and task re-dispatch rate as box plots (box: IQR; line: median; whiskers:  $1.5 \times \text{IQR}$ ; points: individual runs,  $n=15$  per method): AHE’s reactivity costs higher task delay and re-dispatch relative to the commit-once baselines.

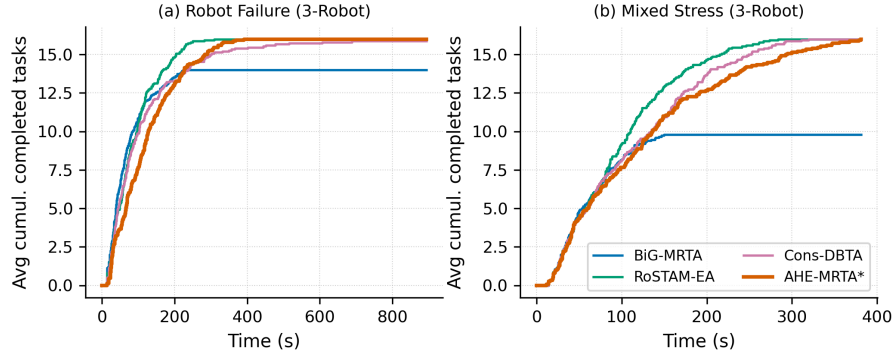


Fig. 12. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: **(a)** *robot\_failure*, **(b)** *mixed\_stress*. AHE-MRTA completes the full task set in both scenarios and tracks the consensus baselines closely, with a modest makespan penalty from re-assignment through Nav2 (*mixed\_stress*: 238 s versus 217 s for Consensus-DBTA and 231 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

cycle.

- **Not when a single regime dominates the whole run and makespan is the objective.** Under uniform deadline pressure at 3 robots, the EDF-style specialists already reach full completion (RoSTAM, Cons-DBTA and AHE all at  $\text{CR} = 1.000$ ), so AHE’s switching yields no meaningful completion edge over them while adding delay; on the strict on-time fitness AHE is within noise of BiG-MRTA at this scale (0.420 vs 0.417). Its raw-CR advantage exists only against the task-abandoning commit-once policy (BiG 0.627 completion). At 10 robots, however, AHE keeps DVR at 0.220 against the specialists’ 0.428–0.504 and attains the highest effective on-time completion (Sec. VI).

### C. Why AHE co-leads rather than dominates the allocation fitness

On the stochastic navigation-proxy fitness (priority-weighted on-time completion), AHE-MRTA and Consensus-DBTA are statistical co-leaders rather than one dominating the other (Table VI). Two mechanisms explain why the strongest methods converge. First, the fitness ceiling is set by *stochastic, position-dependent navigation failure*, not by assignment quality: F58 AHE reaches fitness 1.0 in all nine scale–scenario cells of the perfect-navigation allocation-only model (Table V), and the



TABLE XIV  
EDPS ABLATION (NAV2-INDEPENDENT FITNESS, GEODESIC EXECUTION ORACLE, 5R/25T, 100 SEEDS). ALL VARIANTS FALL WITHIN 0.7 PP MEAN FITNESS; NONE STATISTICALLY SEPARATES FROM EDPS.

| Variant           | Rob. fail. | Deadline | Mix. stress | Mean  |
|-------------------|------------|----------|-------------|-------|
| Full EDPS         | 0.488      | 0.447    | 0.480       | 0.472 |
| fixed: spatial    | 0.479      | 0.448    | 0.484       | 0.471 |
| fixed: edf        | 0.489      | 0.443    | 0.477       | 0.470 |
| fixed: commit     | 0.485      | 0.443    | 0.479       | 0.469 |
| fixed: orphan     | 0.483      | 0.445    | 0.488       | 0.472 |
| fixed: priority   | 0.499      | 0.441    | 0.487       | 0.476 |
| no-override       | 0.489      | 0.447    | 0.478       | 0.471 |
| no-recovery-boost | 0.488      | 0.447    | 0.481       | 0.472 |

spread in the proxy opens only once goals can fail and must be re-attempted. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—so the two converge to the same frontier, while the commit-once BiG-MRTA, which abandons failed tasks, trails. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE’s completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE’s full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware execution ordering could push AHE past the co-leaders: in isolation it lifts deadline fitness by  $\approx 1.3$  pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose co-leading fitness and decisive physical-stack advantages (Sec. VI) are the stronger operating point.

#### D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the EDPS selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure  $\arg \max_i D_i$ ) and the recovery boost (*no-recovery-boost*:  $\delta=0$ ), holding everything else fixed (5r/25t, 100 seeds, `scripts/ablation_edps.py`).

The eight variants span a mean-fitness range of only  $[0.519, 0.528]$  (spread 0.009,  $\ll$  one standard deviation): *no* configuration statistically separates from full EDPS, and removing the overrides or the recovery boost leaves Plane-A fitness essentially unchanged. This is the expected consequence of the saturation identified above (under perfect navigation every method reaches  $\approx 1.0$ ): in the navigation-free plane, allocation quality is not the differentiator, so the switching machinery *cannot* and does not improve it. The value of EDPS is therefore not idealized allocation fitness but the physical-stack behaviour (Sec. VI), where event-triggered recovery and regime- appropriate paradigm selection drive the completion, deadline-robustness, and latency advantages. We report this ablation explicitly rather than omit it: it delimits, honestly, *where* the contribution lies.

#### E. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Figure 13 shows the resulting message footprint stays compact and does not grow with paradigm switching frequency, preserving the low-bandwidth property needed for field deployment.

#### F. Limitations

(i) **Statistical power at the primary scale:** At the 5-robot primary scale ( $n=5$  per cell) no comparison survives Bonferroni correction— $n=5$  is structurally underpowered—whereas at 3 robots ( $n=15$ ) 14 of 63 do; at 10 robots the five seeds per cell are likewise underpowered and no comparison survives correction. We mitigate by reporting 100-seed Nav2-independent fitness as the primary hypothesis test and Cliff’s  $\delta$  as a sample-size-robust effect measure; a larger 10-robot seed budget would add inferential support. (ii) **Recovery-time variance:** Recovery time under *mixed\_stress* carries high variance (10r:  $72 \pm 78$  s) because some seeds have no robot failure within the observation window; the *robot\_failure* scenario, with guaranteed in-window failures, is the cleaner recovery measure. (iii) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-core/15 GB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one; the Nav2-independent plane confirms the allocation policy itself scales further. (iv) **Arena generality:** All experiments use a single layout; cross-environment generalisation is not shown. (v) **Fixed hormone–paradigm mapping:** The  $A$ ,  $S$ , and  $V$  matrices are designer-specified; learning

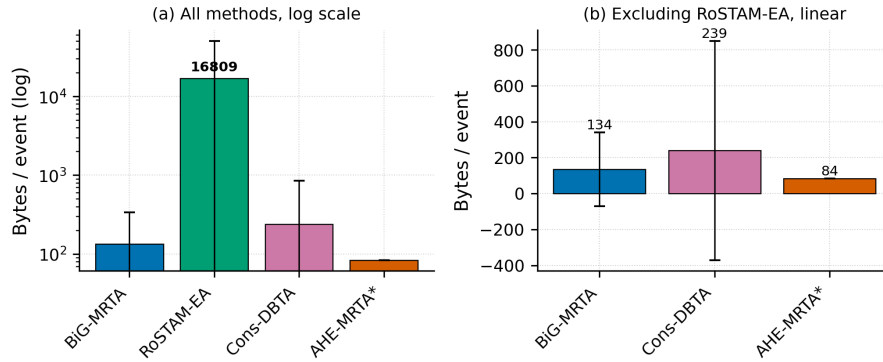


Fig. 13. Communication footprint per allocation event. Only per-robot queues are broadcast; ecosystem internals stay local, keeping bandwidth bounded.

them from domain data is left to future work. (vi) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. (vii) **Ground-truth localization:** The physical benchmark supplies pose through a static map→odom transform at the spawn pose, deliberately isolating allocation quality from localization error. Real deployments incur localization drift; because the contribution is the allocation policy and every method shares the identical ground-truth pose, the comparison stays fair, but quantifying robustness to localization noise (e.g., AMCL) is left to future work. (viii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. We therefore frame EDPS as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector. The ablation in Table XIV (Sec. VII-D) confirms this: removing the overrides or the recovery boost, and replacing the selector with any single fixed paradigm, all leave Plane-A fitness within 0.9 pp of EDPS. The switching machinery’s value is thus confined to the physical stack (Plane B), not the navigation-free plane. (ix) **Allocation-fitness model.** The Plane-A simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the ground-truth Gazebo stack (where Nav2 reaches goals almost always). Plane A therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the physical Plane-B results.

## VIII. CONCLUSION

We presented AHE-MRTA, an ecosystem-driven paradigm-switching framework for online multi-robot task allocation, together with its geodesic, efficiency-guarded F58 extension. Across 100-seed true Nav2-independent allocation-only simulation, F58 AHE holds fitness and CR at 1.000 and DVR at 0.000 across 3/5/10 robots and all three scenarios, with active Jain 0.982–0.992 and decision latency of at most 0.201 ms. In a four-cell seed-matched F45 comparison (3r/5r/10r plus a fresh holdout; 51 of 120 tests significant after Holm) delay falls by 1–2.2 s and geodesic distance by 10–28 m with CR/DVR unchanged, while the active-robot Jain index rises by 0.005–0.029; the only cost is a decision-latency increase of up to 0.12 ms. In the separate stochastic navigation proxy, AHE and the strongest consensus baseline (Consensus-DBTA) are statistical co-leaders. Across a 300-experiment Gazebo benchmark (3r, 5r, and 10r physical scales) with ROS 2 Jazzy and Nav2 on TurtleBot3 robots, AHE-MRTA reaches full completion at the primary 5-robot scale (CR=1.000) and the highest completion rate at 10 robots (CR=0.976–0.988), and leads effective on-time completion under deadline pressure at both scales (0.79 vs  $\leq 0.66$  at 5r, 0.76 vs  $\leq 0.52$  at 10r); the consensus and evolutionary baselines match its completion under deadline pressure only by incurring up to  $2\times$  higher deadline-violation rates (DVR=0.34–0.50 vs AHE’s 0.21–0.22), while under mixed stress and robot failure AHE co-leads—cautious leadership rather than uniform dominance at  $n=5$  per cell. The fairness-guarded repair that delivers this deadline compliance raises decision latency to 16–35 ms across scales—above the sub-millisecond commit-once baselines but within the real-time budget and  $2\text{--}5\times$  below the evolutionary RoSTAM-EA—at the further cost of higher makespan and task delay relative to static-commit consensus methods under mixed-stress conditions. In the matched physical campaign at all three scales (180 runs; 10 common seeds per scenario and scale), F58 (P1R) passes the fairness-guarded promotion gate in seven of nine scale–scenario cells with paired-Wilcoxon support: at the primary scale DVR, delay, and geodesic distance fall, the active-robot Jain index rises, and failure recovery shortens by 34 s; the deadline-pressure DVR and distance gains persist at 10 robots, and the two non-passing cells fail only on criteria whose CIs span zero. This matched campaign doubles as a core-vs-full ablation of AHE-MRTA\*: it isolates the geodesic-ETA fairness repair as the component that lifts AHE from the co-leadership of the idealized plane to its deadline-pressure lead in the physical benchmark. We hope the decomposition into ideal allocation fitness vs. realized Gazebo metrics is useful to other MRTA studies struggling to separate algorithm quality from physical execution noise.

## REFERENCES

- [1] F. Zitouni, R. Maamri, and S. Harous, "Towards a formal analysis of the multi-robot task allocation problem using set theory," *Bulletin of Electrical Engineering and Informatics*, vol. 10, no. 2, pp. 1092–1104, 2021.
- [2] T. Lei, P. Chintam, and C. Luo, "A convex optimization approach to multi-robot task allocation and path planning," *Sensors*, vol. 23, no. 11, p. 5103, 2023.
- [3] H. Aziz, H. Chan, and Á. Cseh, "Multi-robot task allocation – complexity and approximation," *arXiv preprint arXiv:2103.12370*, 2021.
- [4] S. Choudhury, J. K. Gupta, and M. J. Kochenderfer, "Dynamic multi-robot task allocation under uncertainty and temporal constraints," *Autonomous Robots*, vol. 46, no. 1, pp. 231–247, 2021.
- [5] S. Park, Y. D. Zhong, and N. E. Leonard, "Multi-robot task allocation games in dynamically changing environments," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2021, pp. 8678–8684.
- [6] F. Quinton, C. Grand, and C. Lesire, "Communication-preserving bids in market-based task allocation," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2022, pp. 13 708–13 713.
- [7] A. Milot, E. Chauveau, and S. Lacroix, "Market-based multi-robot coordination with HTN planning," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2021, pp. 2606–2612.
- [8] M. Al-Shaboti and U. Baroudi, "Multi-robot task allocation system: Fuzzy auction-based and adaptive multi-threshold approaches," *SN Computer Science*, vol. 2, no. 2, p. 110, 2021.
- [9] P. Ghassemi and S. Chowdhury, "Multi-robot task allocation in disaster response: Addressing dynamic tasks with deadlines and robots with range and payload constraints," *Robotics and Autonomous Systems*, vol. 147, p. 103905, 2022.
- [10] J. García Martín, M. Hanif, and T. Hatanaka, "Predictive receding-horizon multi-robot task allocation with moving tasks," in *Proc. European Control Conference (ECC)*, 2022, pp. 2029–2031, 2023.
- [11] P. Mahato, S. Saha, and C. Sarkar, "Consensus-based fast and energy-efficient multi-robot task allocation," *Robotics and Autonomous Systems*, vol. 159, p. 104270, 2023.
- [12] A. Ozalp, E. Cevikcan, and S. I. Satoglu, "On-line task allocation for multi-robot teams under dynamic scenarios," *Intelligent Decision Technologies*, vol. 18, no. 1, pp. 1–18, 2024.
- [13] A. Khamis, A. Hussein, and A. Elmoghy, "Multi-robot task allocation: A review of the state-of-the-art," in *Cooperative Robots and Sensor Networks*. Springer, 2015, pp. 31–51.
- [14] A. Tariq, H. Masood, S. A. Alsuhbany, and A. Jalal, "Deadline-aware task allocation for heterogeneous robot teams in dynamic environments," *IEEE Access*, vol. 11, pp. 45 821–45 835, 2023.
- [15] V. A. Santos, M. Sarcinelli-Filho, and R. Carelli, "Resilient task allocation for mobile robot teams with fault tolerance via hybrid auction-based mechanism," *Robotics and Autonomous Systems*, vol. 145, p. 103868, 2021.
- [16] B. Peng, X. Zhang, and M. Shang, "A novel competition-based coordination model with dynamic feedback for multi-robot systems," *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 10, pp. 2029–2031, 2023.
- [17] E. K. Burke, M. Hyde, G. Kendall, G. Ochoa, E. Özcan, and J. R. Woodward, "A classification of hyper-heuristic approaches: Revisited," in *Handbook of Metaheuristics*. Springer, 2013, pp. 449–474.
- [18] A. Dutta and S. Bhattacharyya, "Task allocation in multi-agent systems using contract-net protocol with adaptive bid generation," *Intelligent Decision Technologies*, vol. 13, no. 2, pp. 209–219, 2019.
- [19] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, 2009.
- [20] L. Zhang, F. Long, and J. Xiao, "Auction-based online task allocation with deadlines for multi-robot systems," *IEEE Robotics and Automation Letters*, vol. 8, no. 4, pp. 2108–2115, 2023.
- [21] F. Tang, J. Li, and M. Tang, "Context-aware multi-robot task allocation using graph attention networks," *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 4, pp. 2451–2464, 2023.
- [22] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2004, pp. 2149–2154.
- [23] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," in *Science Robotics*, vol. 7, no. 66, 2022, p. eabm6074.
- [24] S. Macenski, F. Martín, R. White, and J. G. Clavero, "The marathon 2: A navigation system," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 2718–2725.
- [25] ROBOTIS, "TurtleBot3 platform for ROS 2 navigation benchmarks," ROBOTIS e-Manual, 2023, [Online; accessed 2026]. [Online]. Available: <https://emanual.robotis.com/docs/en/platform/turtlebot3/overview/>